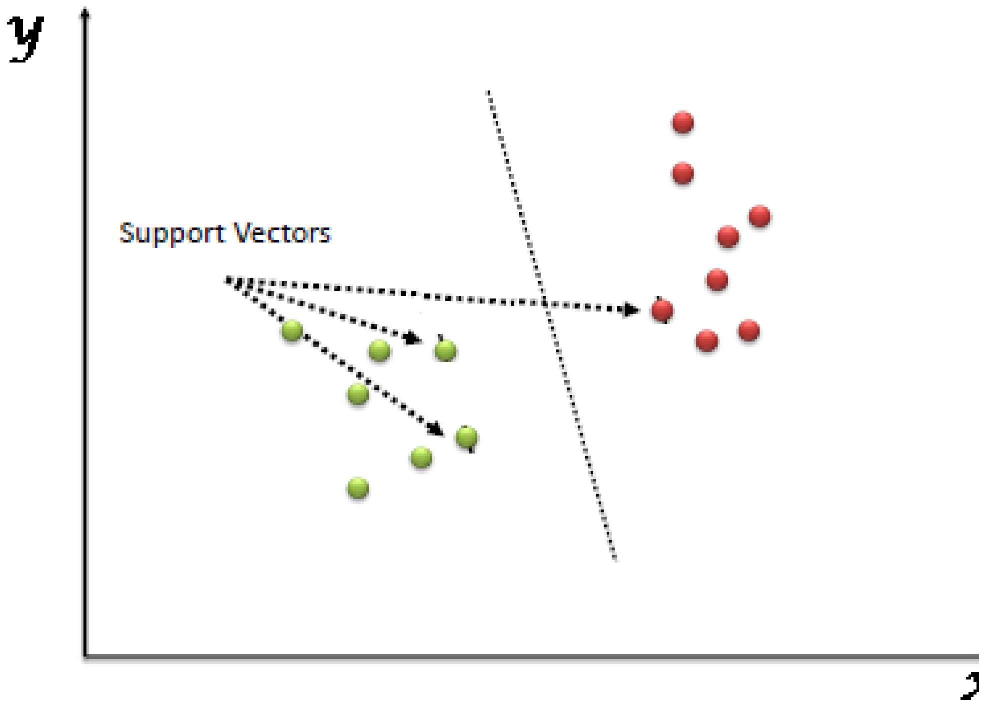
SUPPORT VECTOR MACHINE

SUPPORT VECTOR MACHINE ALGORITHM

- "Support Vector Machine" (SVM) is a supervised machine learning algorithm which can be used for both classification or regression challenges.
- Mostly used in classification problems.
- Linear SVM : A classification technique when training data are linearly separable. (SVC)
- Non-linear SVM : A classification technique when training data are linearly non-separable.
- In the SVM algorithm, we plot each data item as a point in n-dimensional space (where n is number of features you have) with the value of each feature being the value of a particular coordinate. Then, we perform classification by finding the hyper-plane that differentiates the two classes very well.



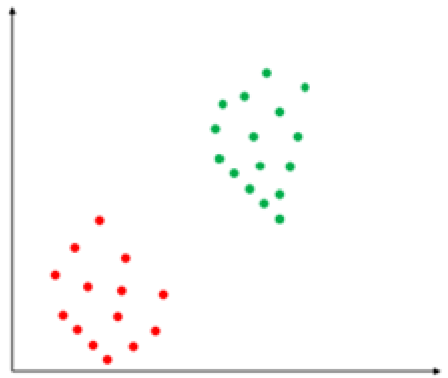
LINEAR SEPARABLE DATA

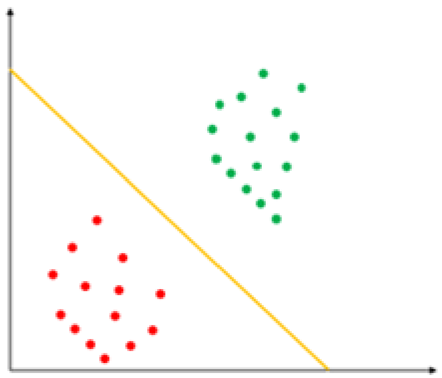
- 1. One-dimensional space

- The data points below the particular point belong to the red class and above that particular point belong to the green class.
- Here the data points are linearly separable in this dimension.

LINEAR SEPARABLE DATA

- 2. Two-dimensional space

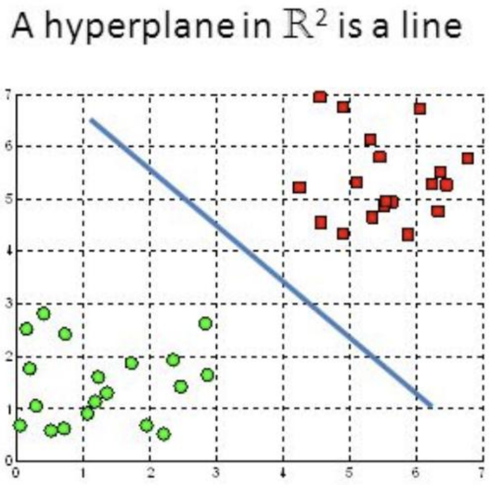


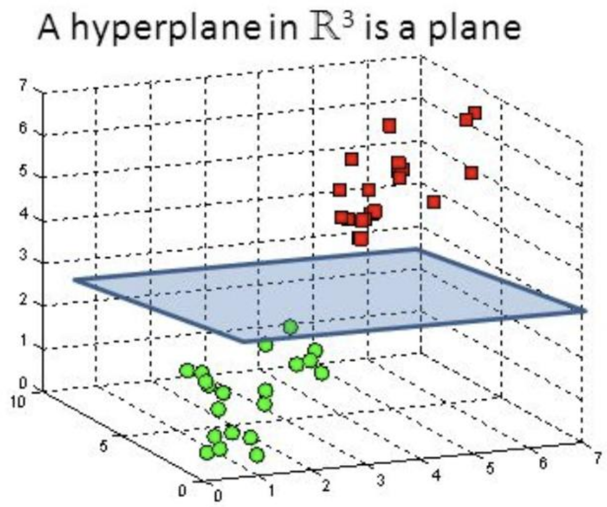


- In two-dimensional space, we can come up with a line that acts as a boundary between two classes.
- Here, the data points are linearly separable in this dimension.

LINEAR SEPARABLE DATA

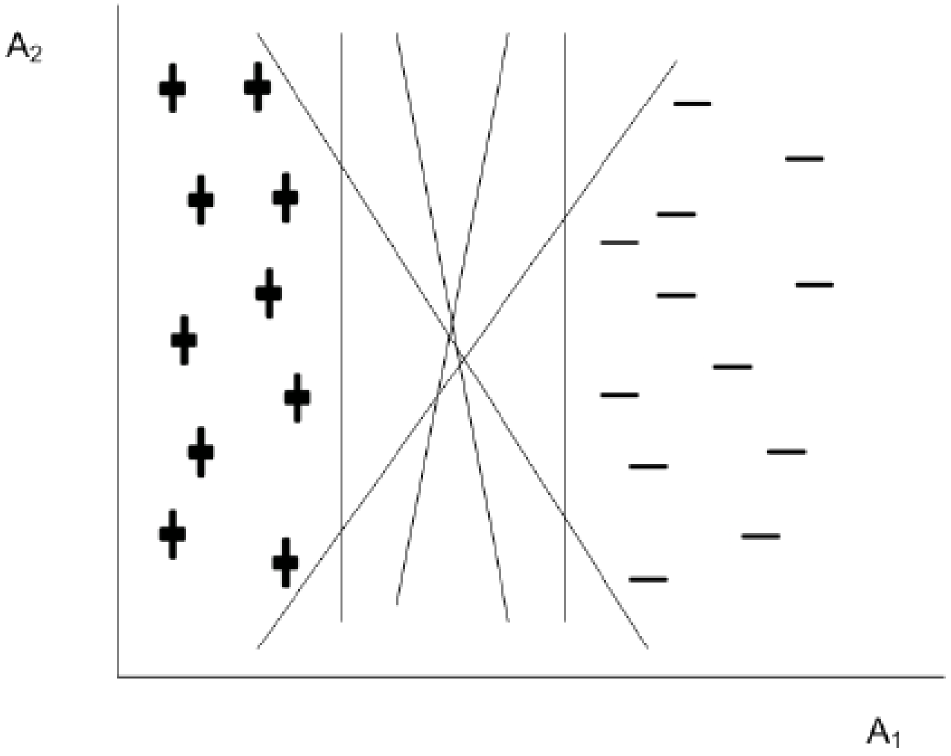
- 3. Three-dimensional space





- In three-dimensional space, we can come up with a two dimensional plane that acts as a boundary between two classes.
- The hyperplane (decision boundary) is a function which is used to differentiate between features.
- A hyperplane is an **n-1** dimensional subspace in an **n**-dimensional space.

A 2D DATA LINEARLY SEPARABLE BY HYPERPLANES

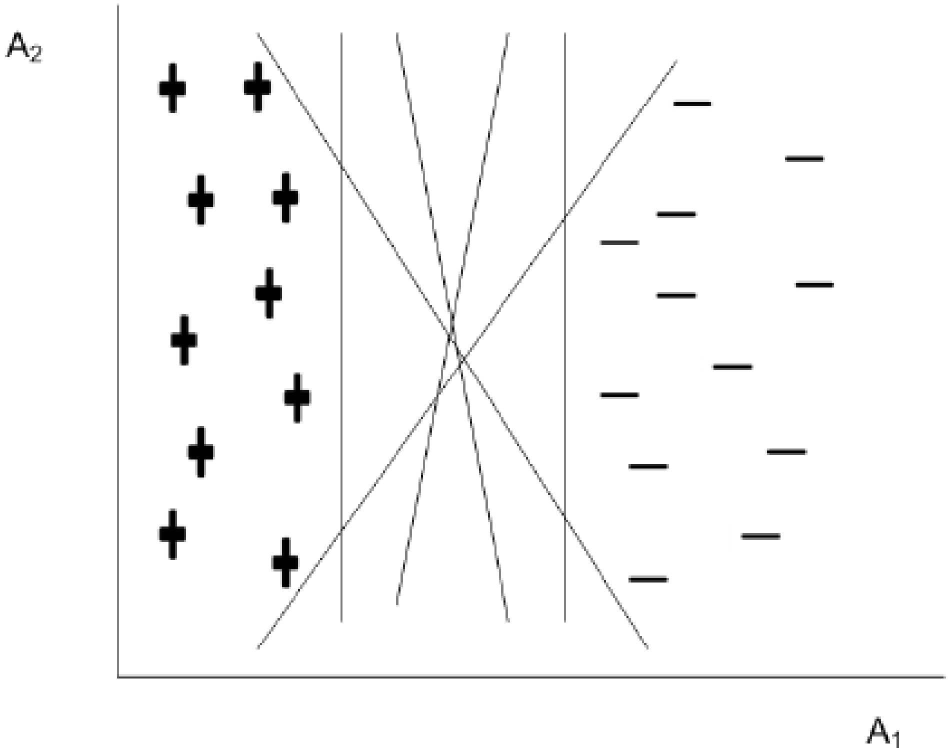


- The data is linearly separable, that is, we can find a hyperplane (in this case, it is a straight line) such that all +'s reside on one side whereas all -'s reside on other side of the hyperplane.
- There are an infinite number of separating lines that can be drawn.

Therefore, the following two questions arise:

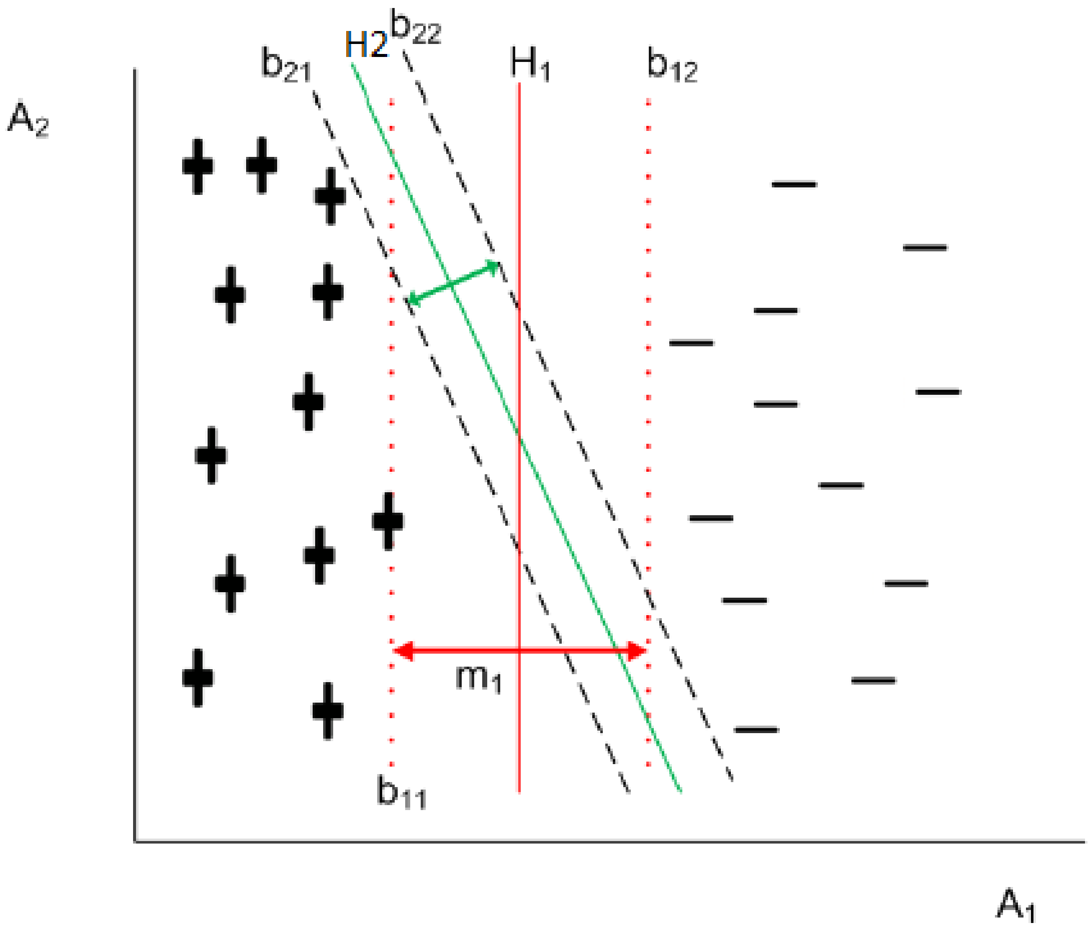
1. Whether all hyperplanes are equivalent?
2. If not, which hyperplane is the best?

A 2D DATA LINEARLY SEPARABLE BY HYPERPLANES



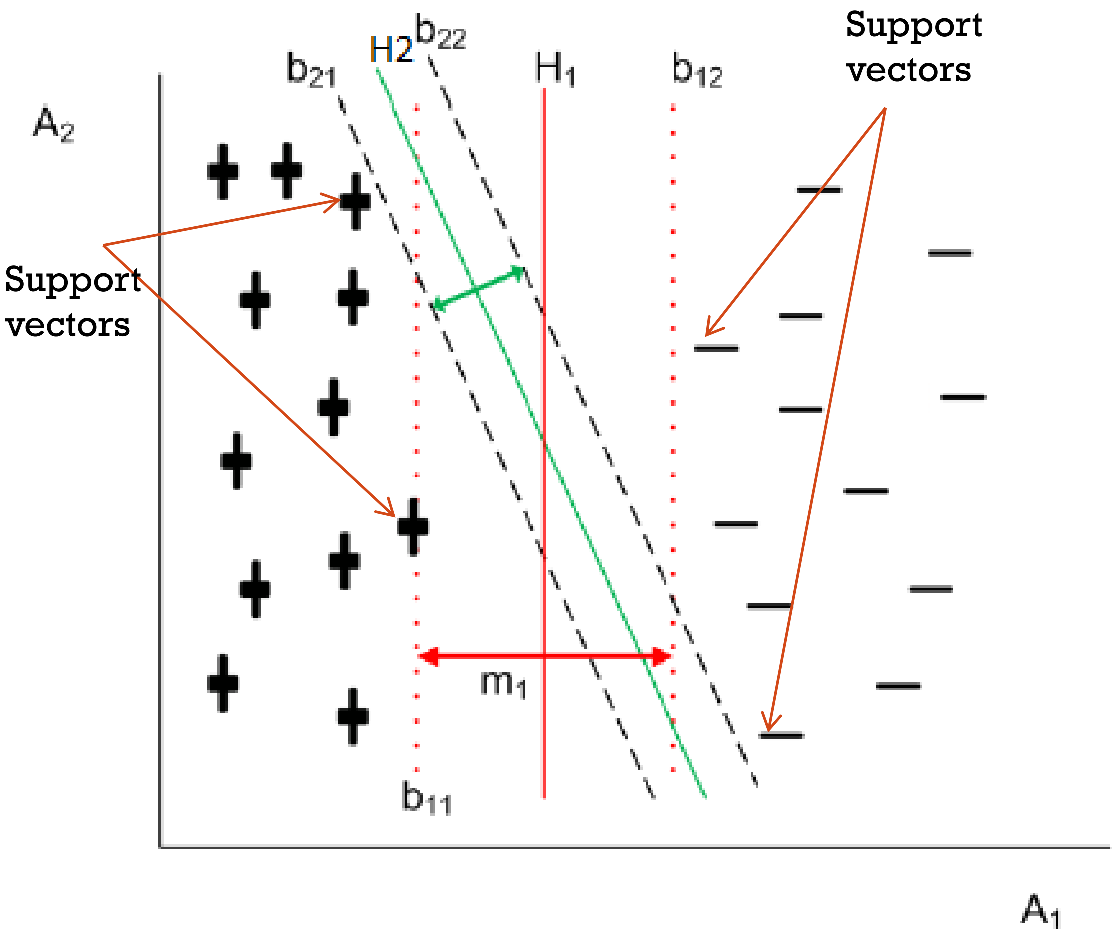
- We may note that so far the classification error is concerned (with training data), all of them are with zero error.
- However, there is no guarantee that all hyperplanes perform equally well on unseen (i.e., test) data.
- Thus, for a good classifier it must choose one of the infinite number of hyperplanes, so that it performs better not only on training data but as well as test data.

MAXIMUM MARGIN HYPERPLANE



- Two hyperplanes H1 and H2 have their own boundaries called decision boundaries(denoted as b11 and b12 for H1 and b21 and b22 for H2).
- A decision boundary is a boundary which is parallel to hyperplane and touches the closest class in one side of the hyperplane. hyperplane is at the middle of its decision boundaries
- The distance between the two decision boundaries of a hyperplane is called the margin. So, if data is classified using Hyperplane H1, then it is with larger margin then using Hyperplane H2.
- The margin of hyperplane implies the error in classifier. In other words, the larger the margin, lower is the classification error.

MAXIMUM MARGIN HYPERPLANE



- Intuitively, the classifier that contains hyperplane with a small margin are more susceptible to model over fitting and tend to classify with weak confidence on unseen data.
- Thus during the training or learning phase, the approach would be to search for the hyperplane with maximum margin.
- Such a hyperplane is called maximum margin hyperplane and abbreviated as MMH.
- This is why a linear SVM is often termed as a **support vector classifier** (SVC) or **maximum margin classifier** (MMC).

WHAT DOES SVM DO?

- SVM algorithm finds the best hyperplane that goes in the middle of the two classes with a maximum margin on both sides.
- Training process is an optimisation
- SVC is based on small number of support vectors

HYPERPLANE(DECISION SURFACE)

- No. of dimensions: M
- The equation of the hyperplane in the 'M' dimension can be given as =

$$\begin{aligned} y &= w_0 + w_1x_1 + w_2x_2 + w_3x_3 \dots \\ &= w_0 + \sum_{i=1}^{m} w_i x_i \\ &= w_0 + w^T X \\ &= b + w^T X \end{aligned}$$

where,

Wi = vectors(W0,W1,W2,W3……Wm)

b = biased term (W0)

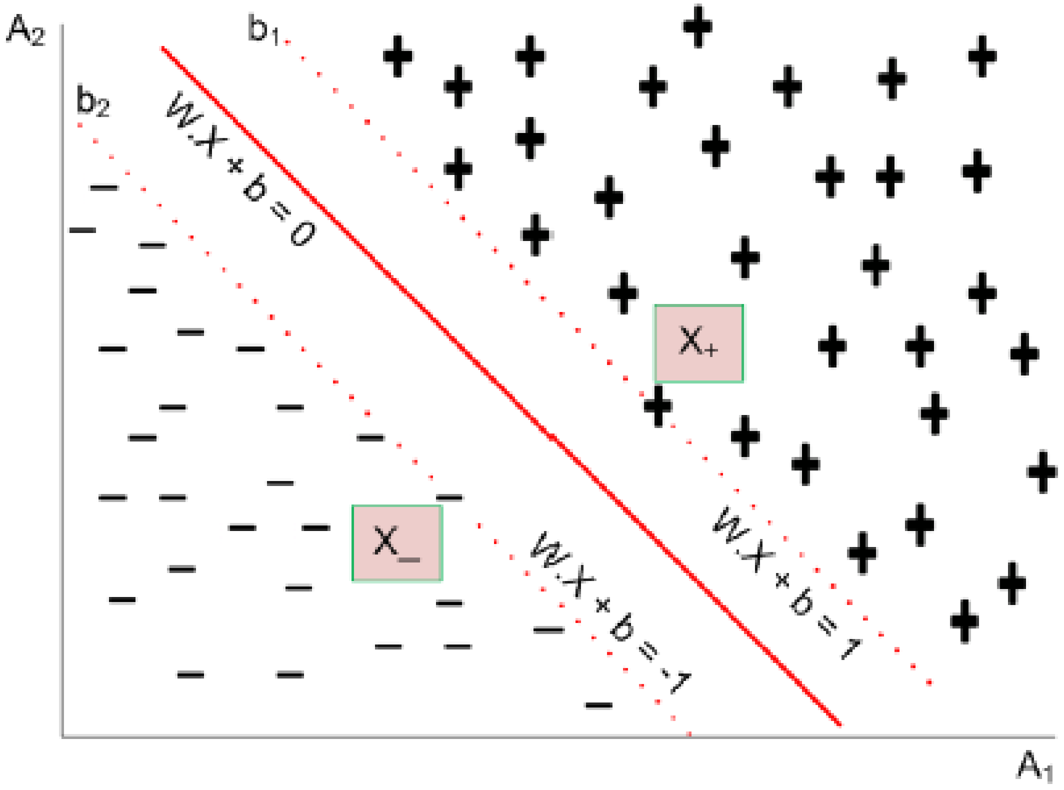
X = variables.

Here, W and b are parameters of the classifier model to be evaluated given a training set D.

FINDING HYPERPLANE

- Let us consider a two-dimensional training set consisting two classes + and - as shown in Fig.
- Suppose, b1 and b2 are two decision boundaries above and belowa hyperplane, respectively.
- Consider any two points X+ and X- as shown in Fig.
- For X+ located above the decision boundary, the equation can be written as

$$W.X_{+} + b = K \qquad \text{where } K > 0$$



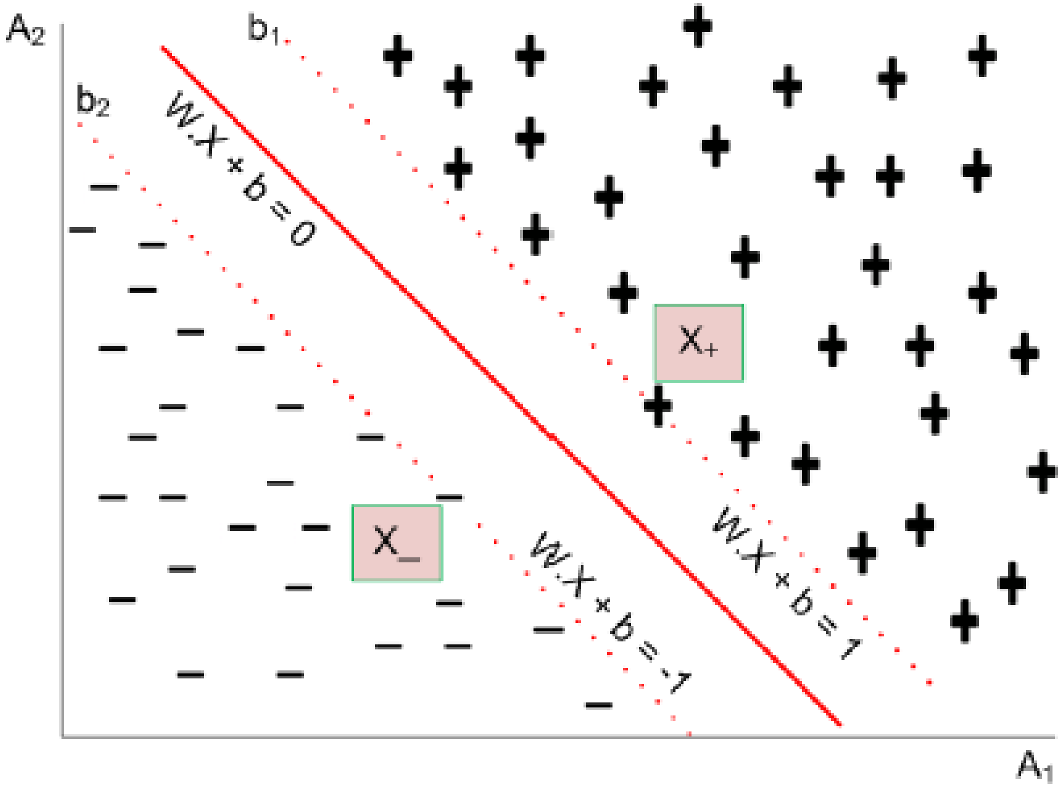
FINDING HYPERPLANE

- Similarly, for any point X_ located below the decision boundary, the equation is

$$W.X_{-} + b = K' \qquad \textit{where } K' < 0$$

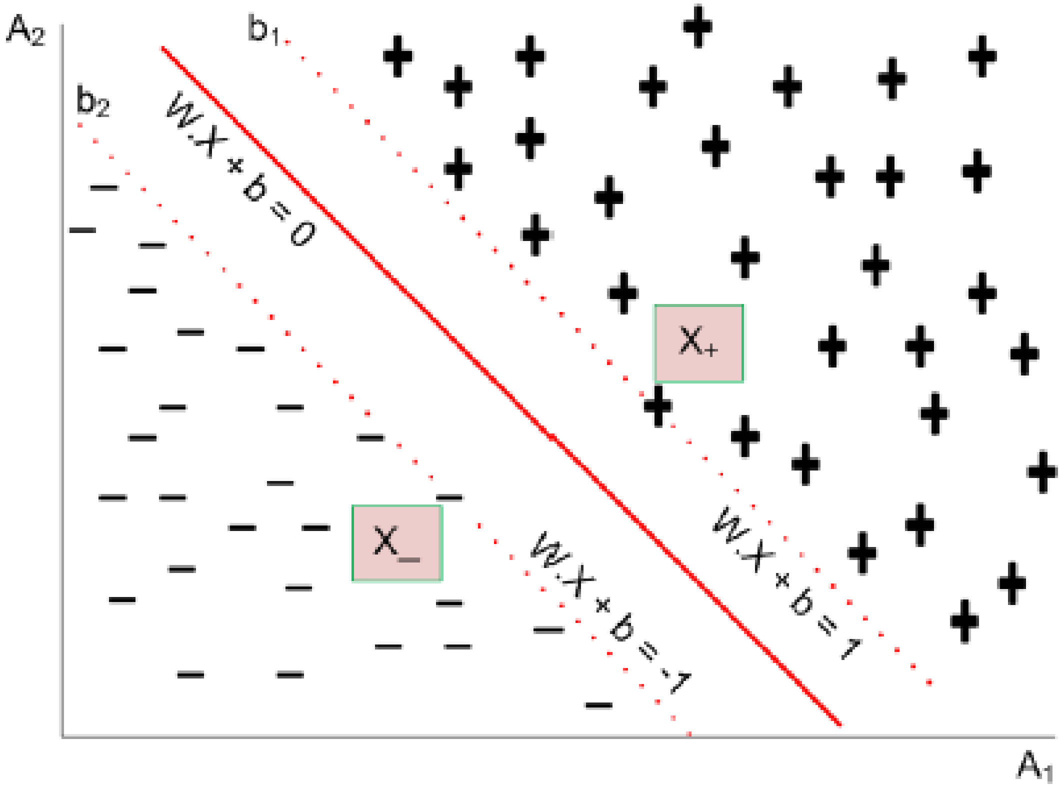
- Thus, if we label all +'s are as class label + and all -'s are class label -, then we can predict the class label Y for any test data X as

$$(Y) = \begin{cases} + & \text{if } W.X + b > 0 \\ - & \text{if } W.X + b < 0 \end{cases}$$



FINDING HYPERPLANE

- Note that W:X + b = 0, the equation representing hyperplane can be interpreted as follows.
- Here, W represents the orientation and b is the intercept of the hyperplane from the origin.
- If both W and b are scaled (up or down) by dividing a non zero constant, we get the same hyperplane.
- This means there can be infinite number of solutions using various scaling factors, all of them geometrical representing the same hyperplane.



FINDING HYPERPLANE

- To avoid such a confusion, we can make W and b unique by adding a constraint that

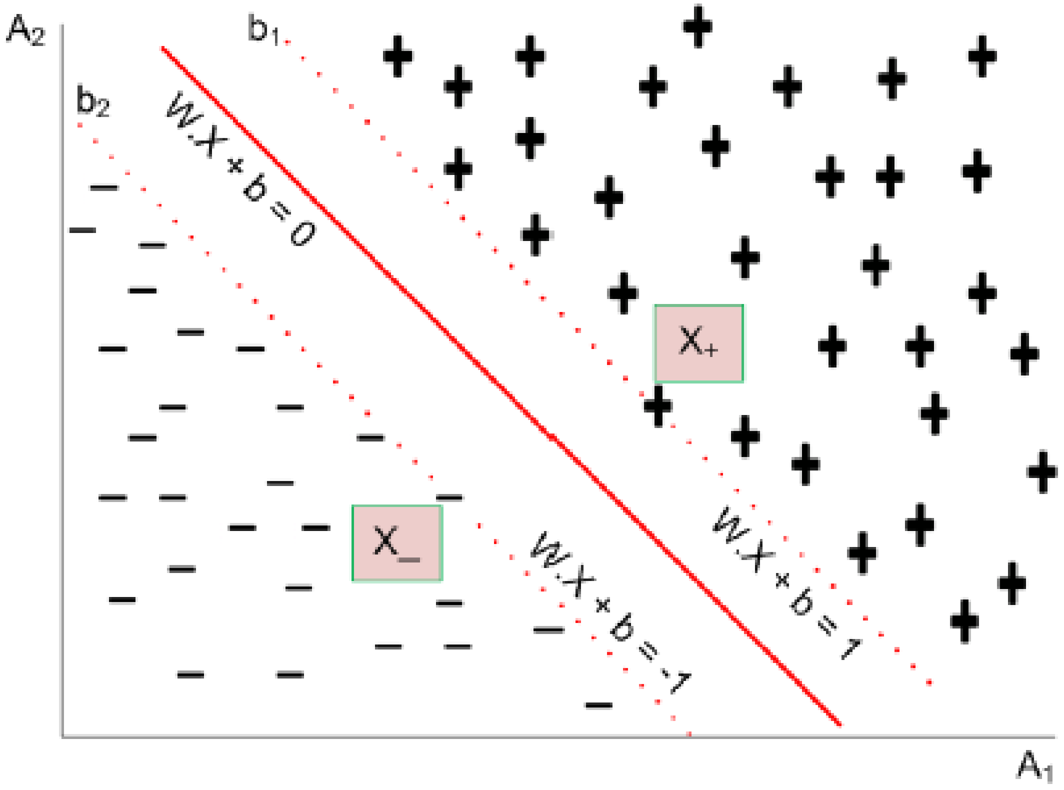
$$W'.X + b' = \pm 1$$

data points on boundary of each class.

- It represents two hyperplane parallel to each other.
- For clarity in notation, we write this as

$$W.X + b = \pm 1.$$

Having this understating, now we are in the position to calculate the margin of a hyperplane.



CALCULATING MARGIN

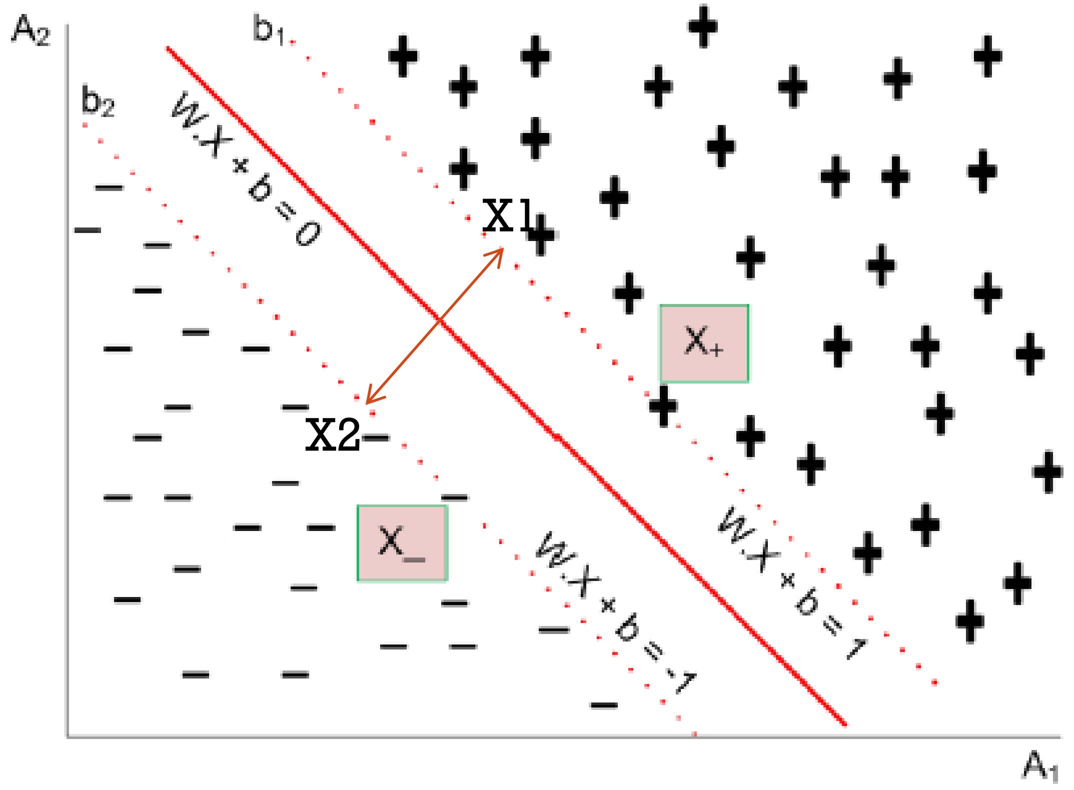
- Suppose, x1 and x2 are two points on the decision boundaries b1 and b2, respectively. Thus,

$$W.X_1 + b = 1$$

and $$W.X_2 + b = -1$$

Or $$W.(x_1 - x_2\,) = 2$$

- This represents a dot (.) product of two vectors W and x1-x2.
- Thus taking magnitude of these vectors, the equation obtained is $\text{margin}(d) = \frac{2}{\|W\|}$
- Where $\|W\| = \sqrt{{w_1}^2 + {w_2}^2 + {w_3}^2 + \cdots + {w_m}^2}$ n m-dimension space.
- In SVM literature, this margin is famously written as $\mu(W, b)$.



TRAINING PHASE OF SVM

- The training phase of SVM involves estimating the parameters W and b for a hyperplane from a given training data.
- The parameters must be chosen in such a way that the following two inequalities are satisfied.

$$W.x_i + b \geq 1 \qquad if \quad y_i = 1$$

$$W.x_i + b \leq -1 \qquad if \quad y_i = -1$$

- These conditions impose the requirements that all training tuples from class Y = + must be located on or above the hyperplane W. x + b = 1, while those instances from class Y = - must be located on or below the hyperplane

W.x + b = -1.

LEARNING FOR A LINEAR SVM

- Both the inequalities $W.x_i + b \geq 1 \quad \text{if} \quad y_i = 1$

$$W.x_i + b \leq -1 \quad \text{if} \quad y_i = -1$$

can be summarized as $y_i(W.x_1 + b) \geq 1 \quad \forall_i i = 1, 2, ..., n$

- The above problem is turned out to be an **optimization problem**, that is, to maximize

$$\mu(W, b) = \frac{2}{\|w\|}$$

SEARCHING FOR MMH

- Maximizing the margin is, however, equivalent to minimizing the following objective function

$$\mu^{'}(W, b) = \frac{||W||^2}{2}$$

- In nutshell, the learning task in SVM, can be formulated as the following **constrained optimization problem.**

$$\text{minimize} \quad \mu^{'}(W, b)$$

$$\text{subject to} \quad y_i(W.x_i + b) \geq 1, \quad i = 1, 2, 3.....n$$

- The above stated constrained optimization problem is popularly known as **convex optimization problem**, where objective function is quadratic and constraints are linear in the parameters W and b.
- The well known technique to solve a convex optimization problem is the **standard Lagrange Multiplier method.**

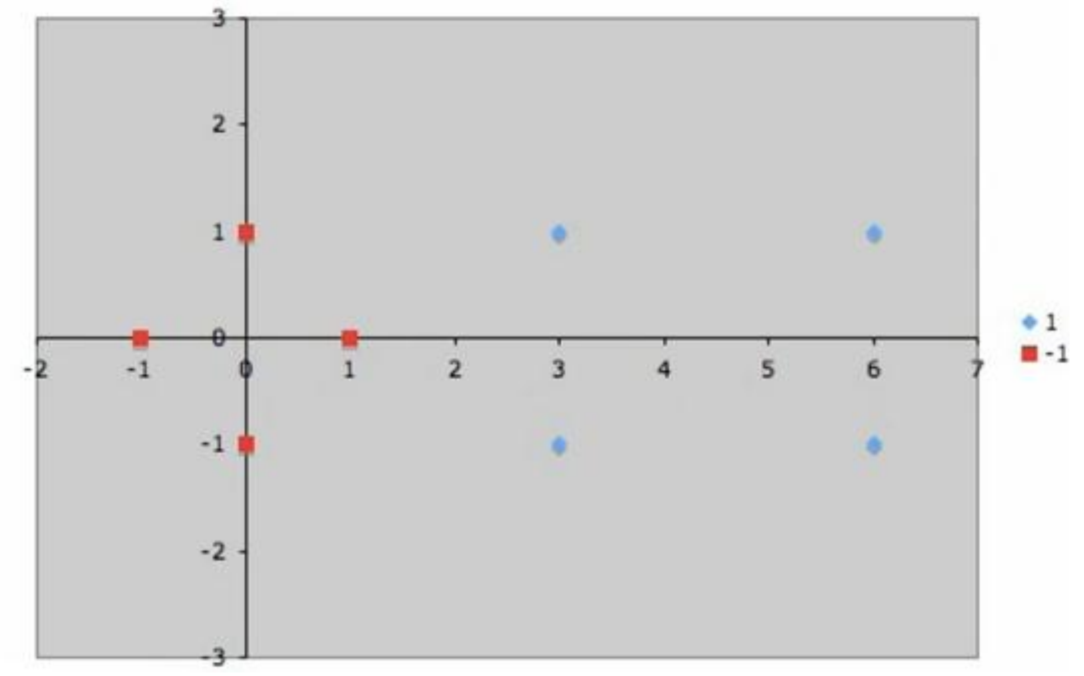
EXAMPLE 1- FINDING HYPERPLANE

Suppose we are given the following positively labeled data points,

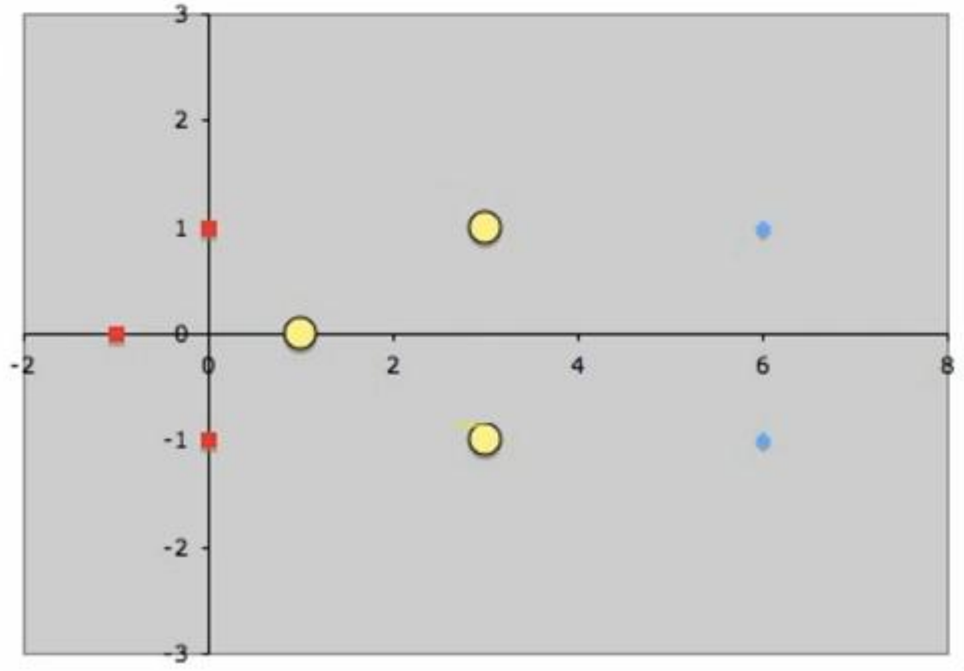
$$\left\{ \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix} \right\}$$

and the following negatively labeled data points,

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right\}$$



EXAMPLE 1- FINDING HYPERPLANE



- Each vector is augmented with a 1 as a bias input
- So, $s_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, then $\tilde{s_1} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$
- Similarly,
- $s_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, then $\tilde{s_2} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$ and $s_3 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$, then $\tilde{s_3} = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$

EXAMPLE 1- FINDING HYPERPLANE

$$\begin{aligned}\alpha_1 \tilde{s_1}\cdot\tilde{s_1} + \alpha_2\tilde{s_2}\cdot\tilde{s_1} + \alpha_3\tilde{s_3}\cdot\tilde{s_1} &= -1\\ \alpha_1 \tilde{s_1}\cdot\tilde{s_2} + \alpha_2\tilde{s_2}\cdot\tilde{s_2} + \alpha_3\tilde{s_3}\cdot\tilde{s_2} &= +1\\ \alpha_1 \tilde{s_1}\cdot\tilde{s_3} + \alpha_2\tilde{s_2}\cdot\tilde{s_3} + \alpha_3\tilde{s_3}\cdot\tilde{s_3} &= +1\end{aligned}$$

$$\alpha_1(1+0+1)+\alpha_2(3+0+1)+\alpha_3(3+0+1) = -1$$
$$\alpha_1(3+0+1)+\alpha_2(9+1+1)+\alpha_3(9-1+1) = 1$$
$$\alpha_1(3+0+1)+\alpha_2(9-1+1)+\alpha_3(9+1+1) = 1$$

$$\alpha_1\begin{pmatrix}1\\0\\1\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix}+\alpha_2\begin{pmatrix}3\\1\\1\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix}+\alpha_3\begin{pmatrix}3\\-1\\1\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix} = -1$$

$$\alpha_1\begin{pmatrix}1\\0\\1\end{pmatrix}\begin{pmatrix}3\\1\\1\end{pmatrix}+\alpha_2\begin{pmatrix}3\\1\\1\end{pmatrix}\begin{pmatrix}3\\1\\1\end{pmatrix}+\alpha_3\begin{pmatrix}3\\-1\\1\end{pmatrix}\begin{pmatrix}3\\1\\1\end{pmatrix} = 1$$

$$\alpha_1\begin{pmatrix}1\\0\\1\end{pmatrix}\begin{pmatrix}3\\-1\\1\end{pmatrix}+\alpha_2\begin{pmatrix}3\\1\\1\end{pmatrix}\begin{pmatrix}3\\-1\\1\end{pmatrix}+\alpha_3\begin{pmatrix}3\\-1\\1\end{pmatrix}\begin{pmatrix}3\\-1\\1\end{pmatrix} = 1$$

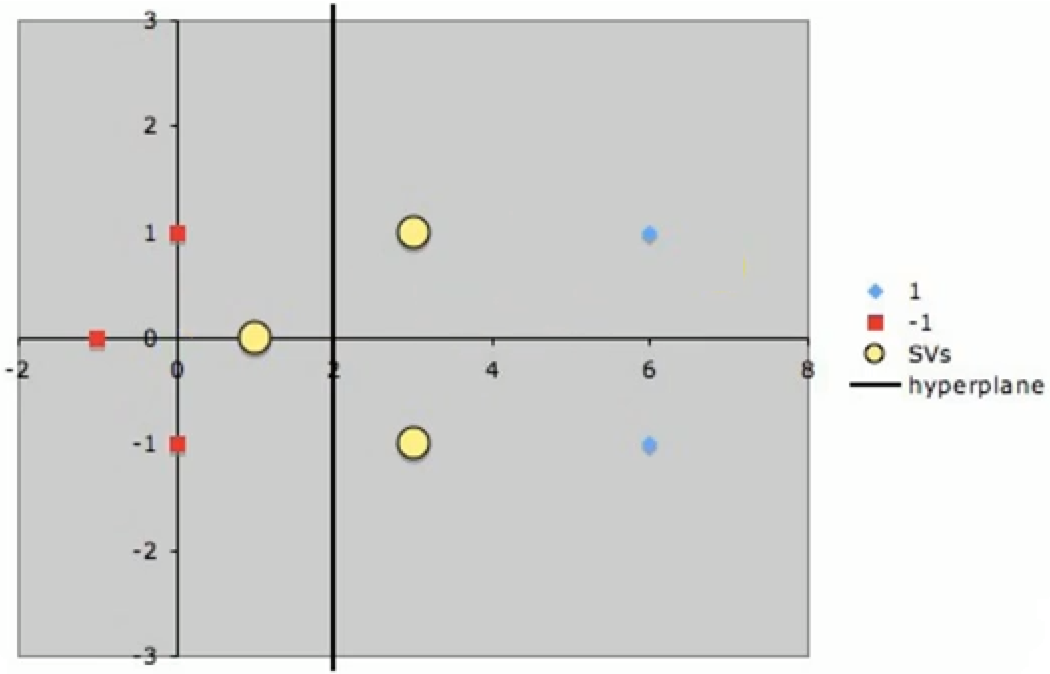
$$2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1$$
$$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1$$
$$4\alpha_1 + 9\alpha_2 + 11\alpha_3 = 1$$

$$\alpha_1 = -3.5$$
$$\alpha_2 = 0.75$$
$$\alpha_3 = 0.75$$

EXAMPLE 1- FINDING HYPERPLANE

$$\begin{aligned}\tilde{w} &= \sum_i \alpha_i \tilde{s}_i \\ &= -3.5\begin{pmatrix}1\\0\\1\end{pmatrix} + 0.75\begin{pmatrix}3\\1\\1\end{pmatrix} + 0.75\begin{pmatrix}3\\-1\\1\end{pmatrix} \\ &= \begin{pmatrix}1\\0\\-2\end{pmatrix}\end{aligned}$$

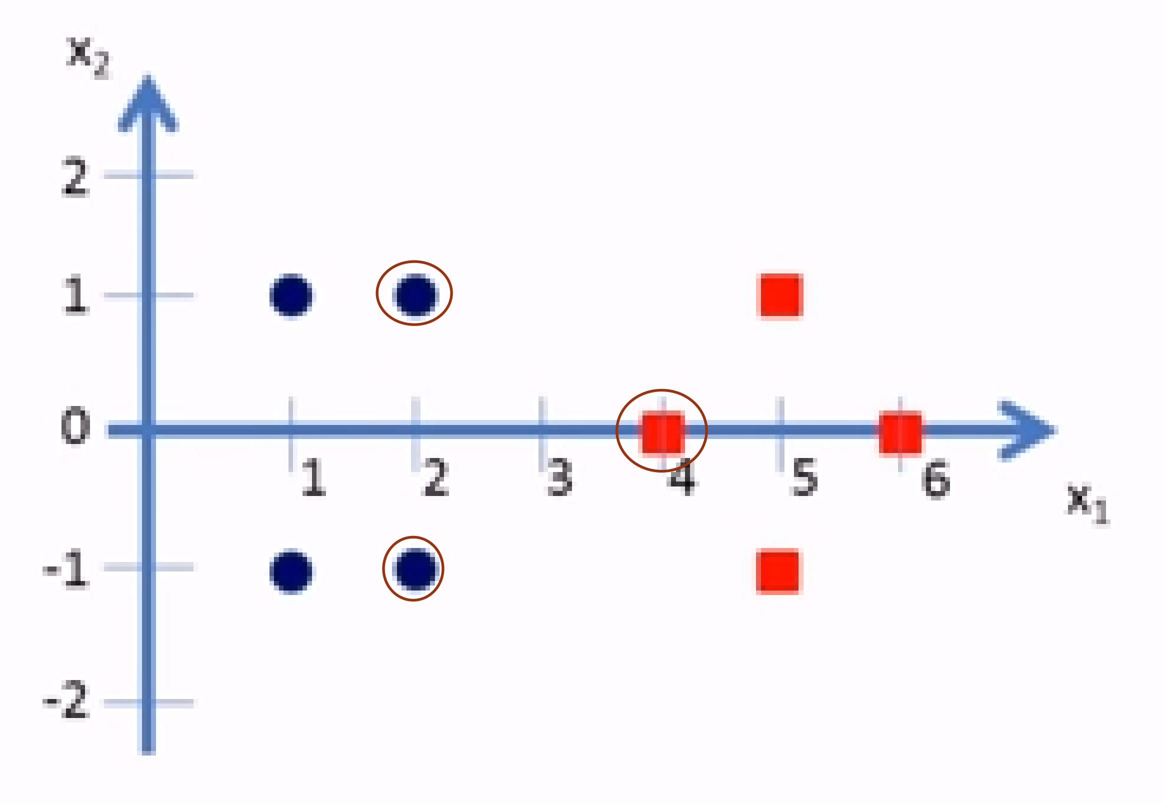
- Finally, remembering that our vectors are augmented with a bias.
- We can equate the last entry in $\widetilde{w}$ as the hyperplane offset b and write the separating
- Hyperplane equation $\boldsymbol{y} = \boldsymbol{wx} + \boldsymbol{b}$
- with $\boldsymbol{w} = \binom{1}{0}$ and $\boldsymbol{b} = -\mathbf{2}$.



EXAMPLE 2-FINDING HYPERPLANE

- Given data points:

x1	x2	y
1	1	-1
2	1	-1
1	-1	-1
2	-1	-1
4	0	+1
5	1	+1
6	0	+1
5	-1	+1



EXAMPLE 2-FINDING HYPERPLANE

$$S_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$S_2 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$S_3 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

$$\widetilde{S_1} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

$$\widetilde{S_2} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

$$\widetilde{S_3} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$$

$$\alpha_1 \widetilde{S_1}.\widetilde{S_1} + \alpha_2 \widetilde{S_2}.\widetilde{S_1} + \alpha_3 \widetilde{S_3}.\widetilde{S_1} = -1 \ (-ve\ class)$$

$$\alpha_1 \widetilde{S_1}.\widetilde{S_2} + \alpha_2 \widetilde{S_2}.\widetilde{S_2} + \alpha_3 \widetilde{S_3}.\widetilde{S_2} = -1 \ (-ve\ class)$$

$$\alpha_1 \widetilde{S_1}.\widetilde{S_3} + \alpha_2 \widetilde{S_2}.\widetilde{S_3} + \alpha_3 \widetilde{S_3}.\widetilde{S_3} = +1 \ (+ve\ class)$$

EXAMPLE 2-FINDING HYPERPLANE

$$\tilde{S_1} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \quad \tilde{S_2} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad \tilde{S_3} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$$

$$\alpha_1 \widetilde{S_1}.\widetilde{S_1} + \alpha_2 \widetilde{S_2}.\widetilde{S_1} + \alpha_3 \widetilde{S_3}.\widetilde{S_1} = -1 \ (-ve\ class)$$

$$\alpha_1 \widetilde{S_1}.\widetilde{S_2} + \alpha_2 \widetilde{S_2}.\widetilde{S_2} + \alpha_3 \widetilde{S_3}.\widetilde{S_2} = -1 \ (-ve\ class)$$

$$\alpha_1 \widetilde{S_1}.\widetilde{S_3} + \alpha_2 \widetilde{S_2}.\widetilde{S_3} + \alpha_3 \widetilde{S_3}.\widetilde{S_3} = +1 \ (+ve\ class)$$

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} . \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} . \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} . \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} . \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} . \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} . \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} . \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} . \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} . \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} = +1$$

After simplification we get:

$$6\alpha_1 + 4\alpha_2 + 9\alpha_3 = -1$$

$$4\alpha_1 + 6\alpha_2 + 9\alpha_3 = -1$$

$$9\alpha_1 + 9\alpha_2 + 17\alpha_3 = +1$$

Simplifying the above 3 simultaneous equations we get: α_1=α_2= -3.25 and α_3= 3.5.

EXAMPLE 2-FINDING HYPERPLANE

The hyperplane that discriminates the positive class from negative class is given by

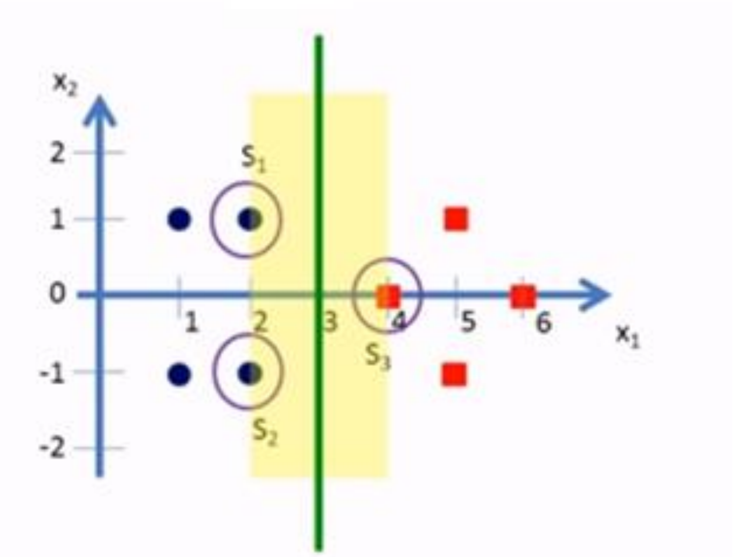
$$\bar{W} = \sum_i \alpha_i \bar{s}_i$$

- Substituting the values we get:

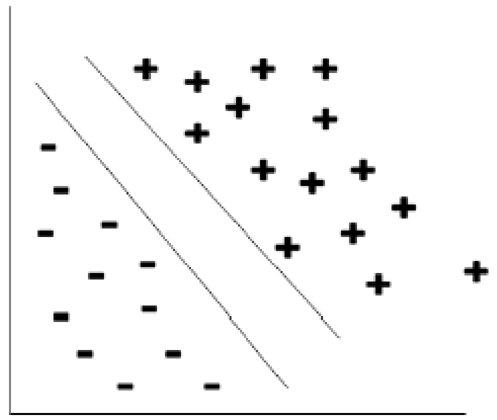
$$\tilde{w} = \alpha_1 \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$$

$$\tilde{w} = (-3.25).\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + (-3.25).\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + (3.5).\begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$$

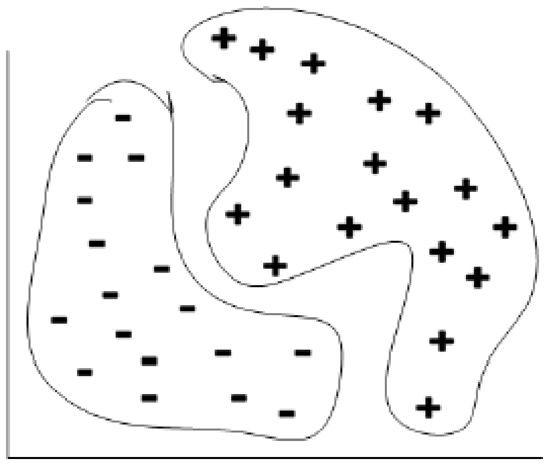
- Our vectors are augmented with a bias.
- Hence we can equate the entry in $\tilde{w}$ as the hyper plane with an offset b.
- Therefore the separating hyper plane equation $y = wx + b$ with $w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and offset $b = -3$.



NON-LINEAR SVM



(a) Linearly separable

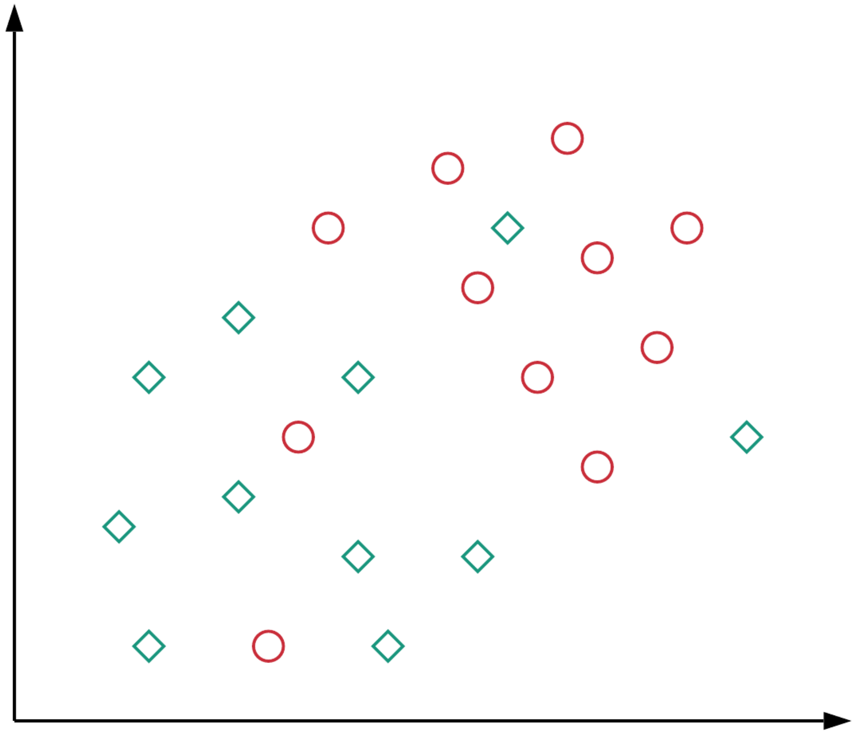


(b) Linearly non-separable

Such a linearly not separable data can be classified using two approaches.

1. Linear SVM with soft margin
2. Non-linear SVM

LINEAR INSEPARABILITY

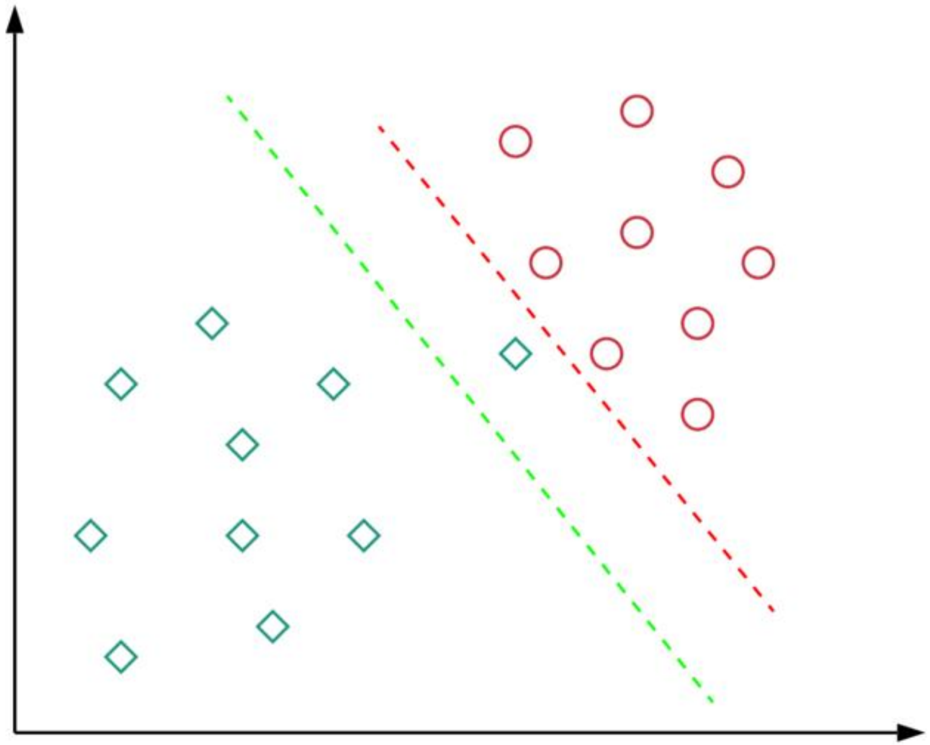


- There's no specific linear decision boundary that can perfectly separate the data, i.e. *the data is linearly inseparable*.
- This can be attributed to the fact that usually, the features we derive from the data don't contain *sufficient information* so that we can clearly separate the two classes.
- This is usually the case in many real-world applications.
- Hard margin : No feasible solution exists, $\alpha = \infty$ for violating points

Solution:

1. Soft margin SVM
2. Kernel Trick

SOFT MARGIN SVM



Which decision boundary is better? Red or Green?

The red decision boundary perfectly separates all the training points.

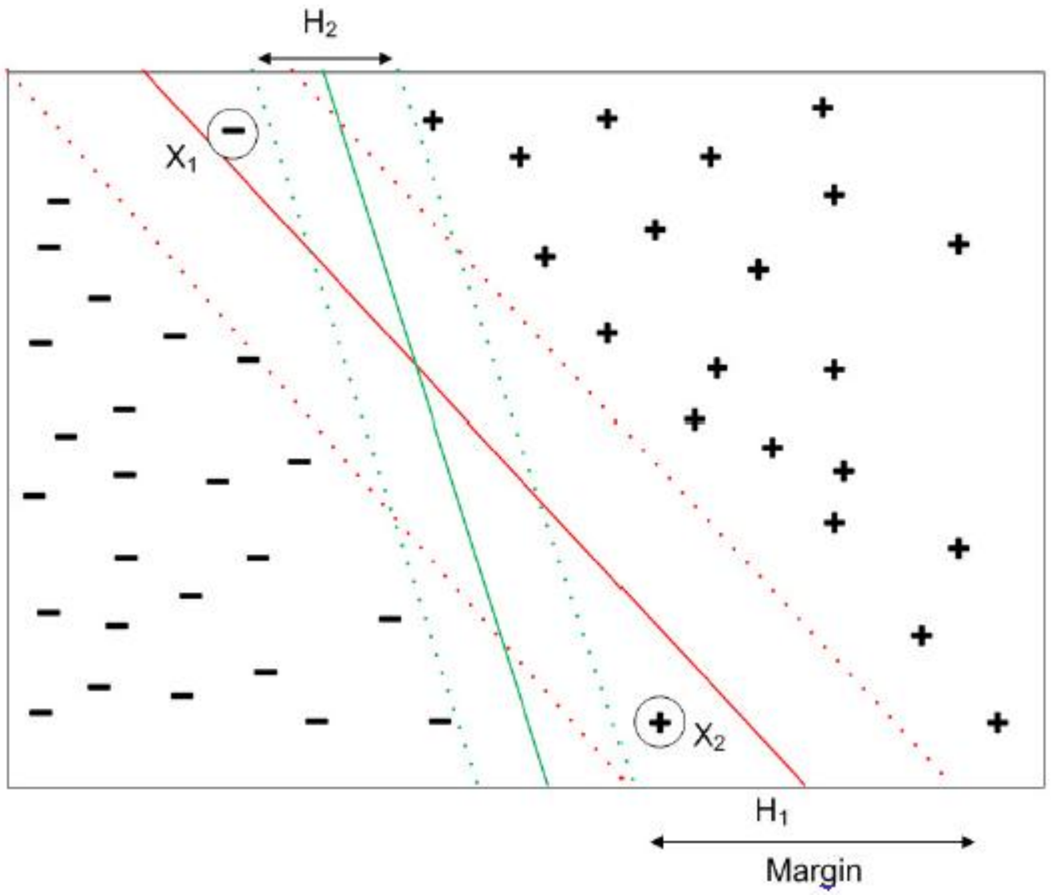
Is it really a good idea of having a decision boundary with such less margin?

Do you think such kind of decision boundary will generalize well on unseen data?

The answer is: No.

The green decision boundary has a wider margin that would allow it to generalize well on unseen data. In that sense, soft margin formulation would also help in avoiding the overfitting problem.

SOFT MARGIN SVM



- Suppose, X1 and X2 are two instances.
- We see that the hyperplane H1 classifies wrongly both X1 and X2.
- Also, we may note that with X1 and X2, we could draw another hyperplane namely H2, which could classify all training data correctly.
- However, H1 is more preferable than H2 as H1 has higher margin compared to H2 and thus H1 is less susceptible to over fitting.

SOFT MARGIN SVM

Concept:

Allow SVM to make a certain number of mistakes and keep margin as wide as possible so that other points can still be classified correctly. This can be done simply by modifying the objective of SVM.

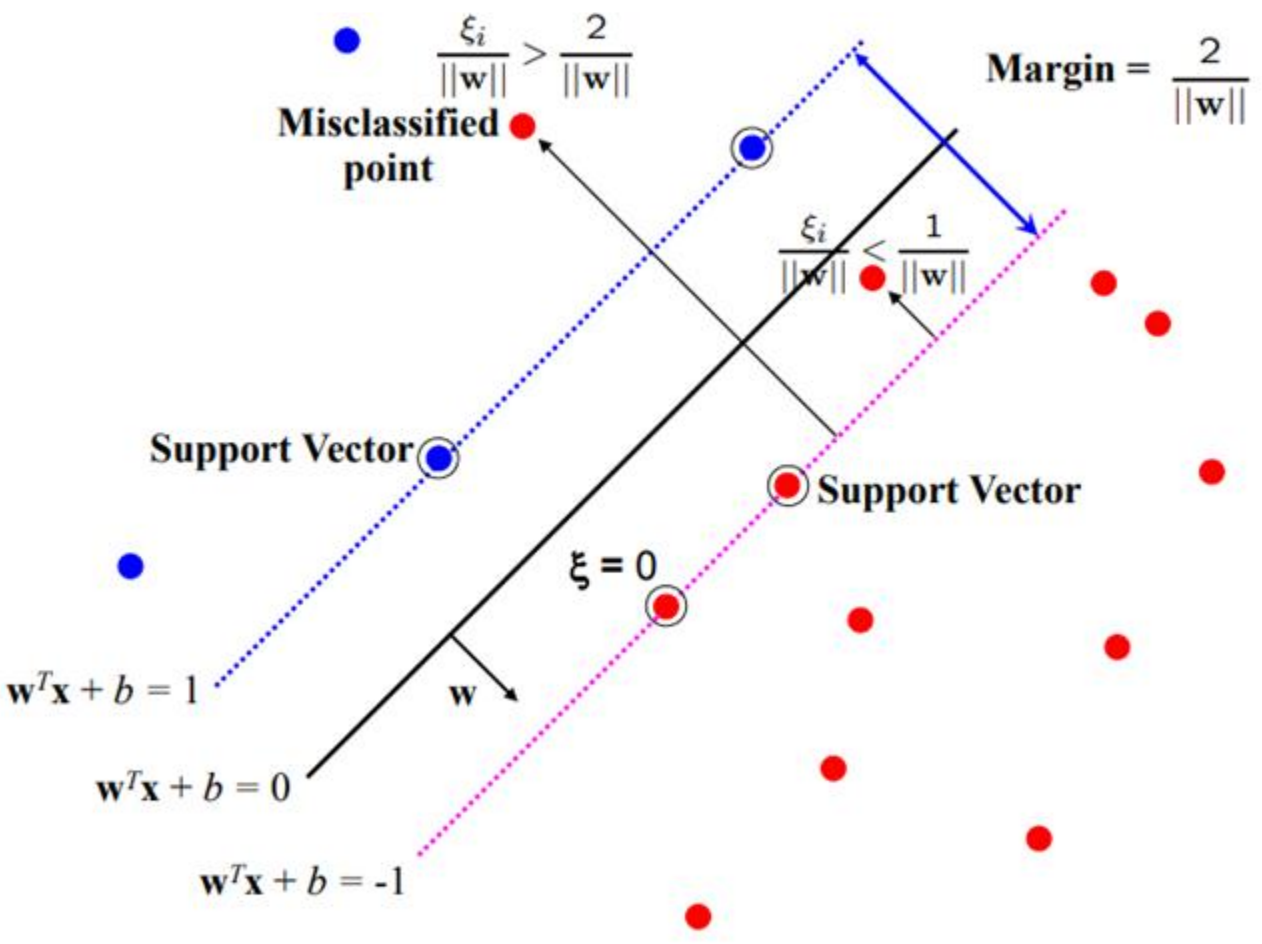
HOW IT WORKS (MATHEMATICALLY)?

- Objective of soft margin SVM

$$L = \frac{1}{2}||w||^2 + C(\#\ of\ mistakes)$$

- This equation differs from the original objective in the second term. Here, **C** is a hyperparameter that decides the trade-off between maximizing the margin and minimizing the mistakes. When **C** is small, classification mistakes are given less importance and focus is more on maximizing the margin, whereas when **C** is large, the focus is more on avoiding misclassification at the expense of keeping the margin small.

"SLACK" VARIABLES



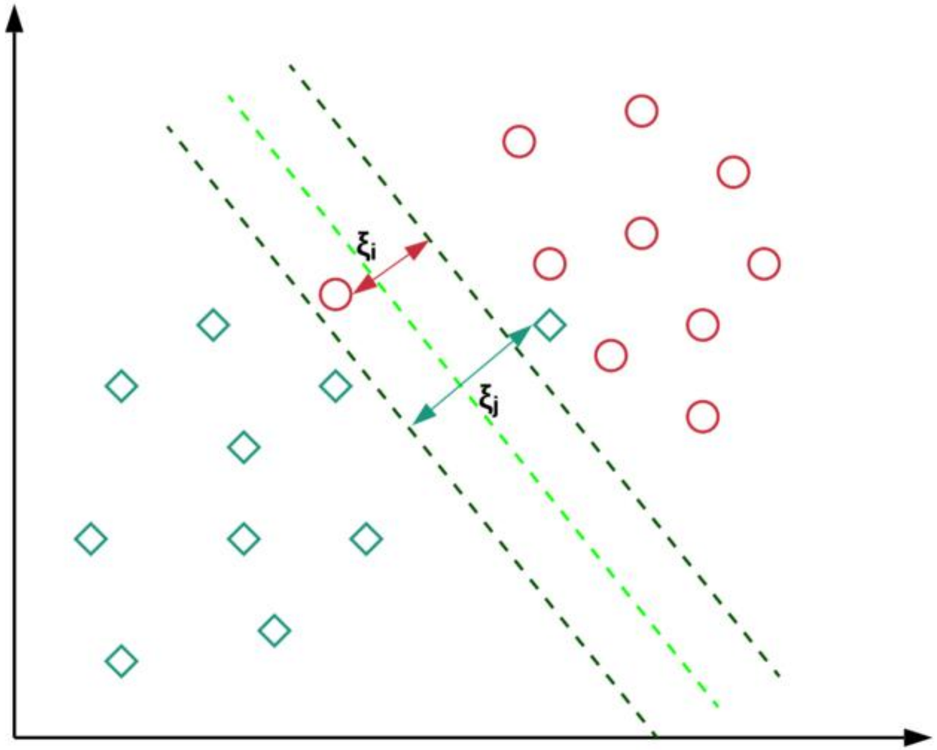
$\xi_i \geq 0$

- for $0 < \xi \leq 1$ point is between margin and correct side of hyper-plane. This is a **margin violation**
- for $\xi > 1$ point is **misclassified**

- Not all mistakes are equal. Data points that are far away on the wrong side of the decision boundary should incur more penalty as compared to the ones that are closer.

- For every data point **x_i**, we introduce a slack variable ξ_**i**. The value of ξ_**i** is the distance of **x_i** from the *corresponding class's margin* if **x_i** is on the wrong side of the margin, otherwise zero. Thus the points that are far away from the margin on the wrong side would get more penalty.

SOFT MARGIN SVM



$$L = \frac{1}{2}\|w\|^2 + C(\#\ of\ mistakes)$$

The modified objective function can be written as

$$f(W) = \frac{||W||^2}{2} + c.\sum_{i=1}^{n}(\xi_i)^{\phi}$$

where c and ϕ are user specified parameters representing the penalty of misclassifying the training data.

Usually, $\phi = 1$. The larger value of c implies more penalty.

SOFT MARGIN SVM-CONSTRAINT

- Each data point **x_i** needs to satisfy the following constraint:

$$y_i(\vec{w}.\vec{x}_i + b) \geq 1 - \xi_i$$

Here, the left-hand side of the inequality could be thought of like the confidence of classification.

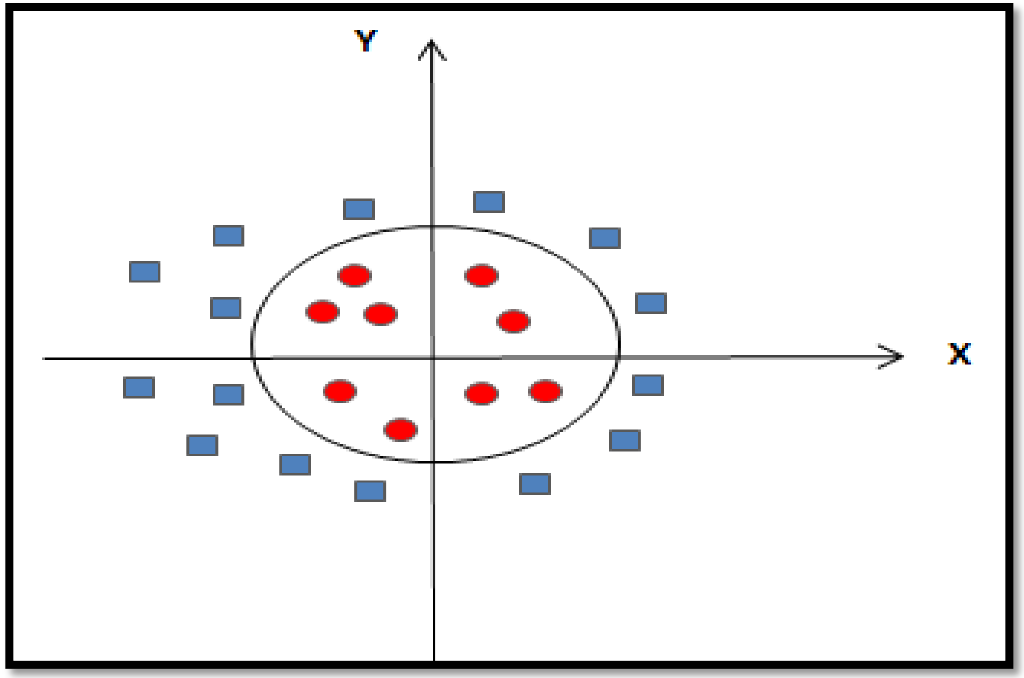
Confidence score ≥ 1 : classifier has classified the point correctly.

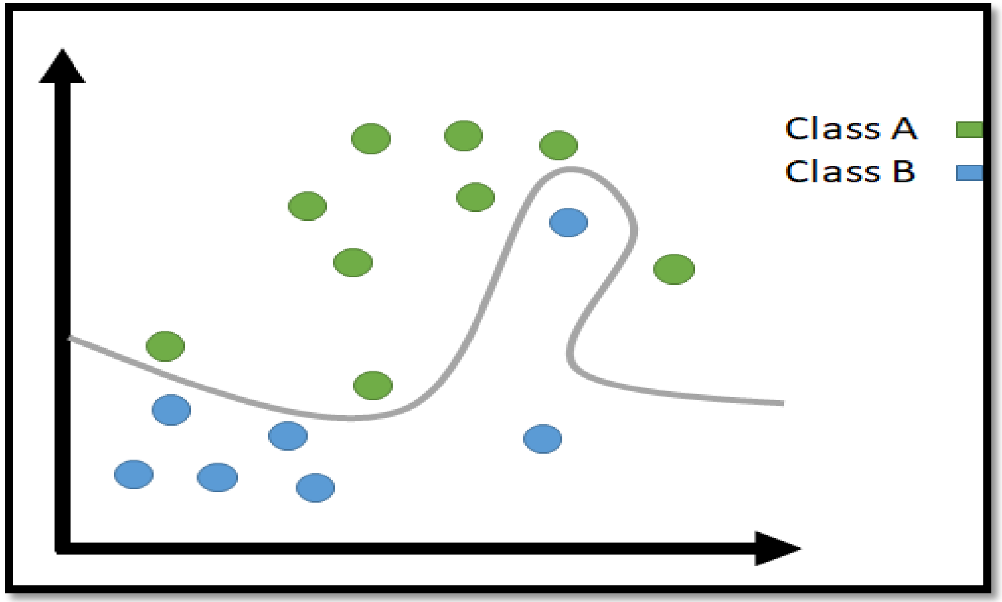
Confidence score ≤ 1 : classifier did not classify the point correctly and incurring a linear penalty of **ξ_i**.

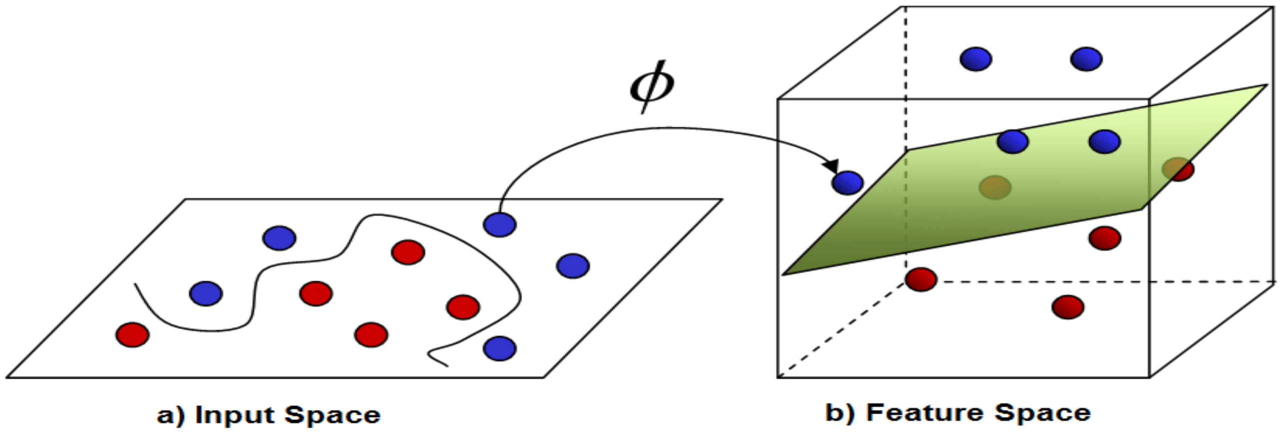
NON LINEAR SVM

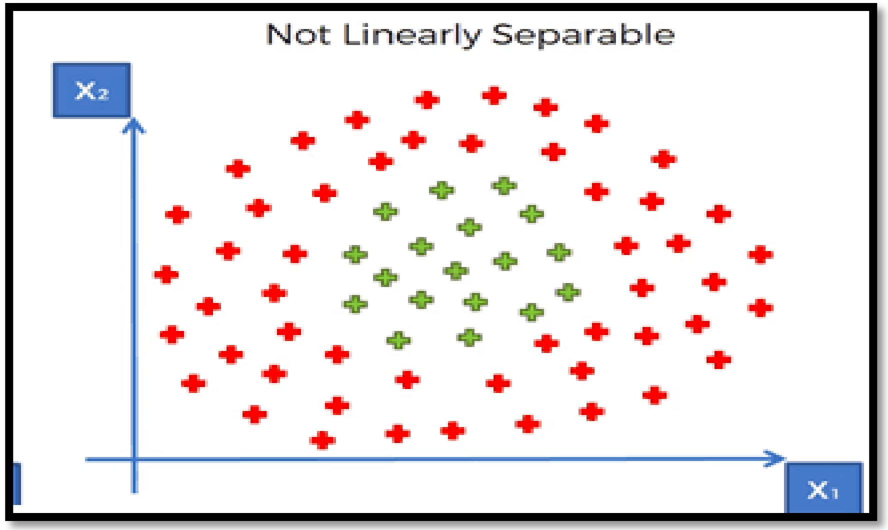
- If **data is linearly arranged**, then we can **separate it by using a straight line**, but for **non-linear data,** we **cannot draw a single straight line.**
- **Imagine a case** - if there is no straight line (or hyper plane) which can separate two classes? In the **image shown below**, there is a **circle in 2D with red and blue data points** all over it such that **adjacent data points are of different colors.**

CONTD..


Y
X


Class A
Class B

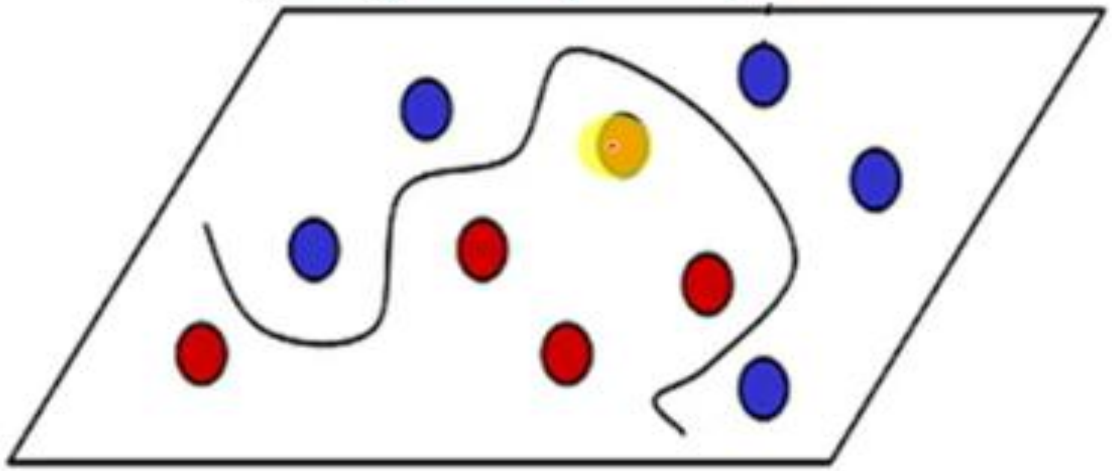

ϕ
a) Input Space
b) Feature Space

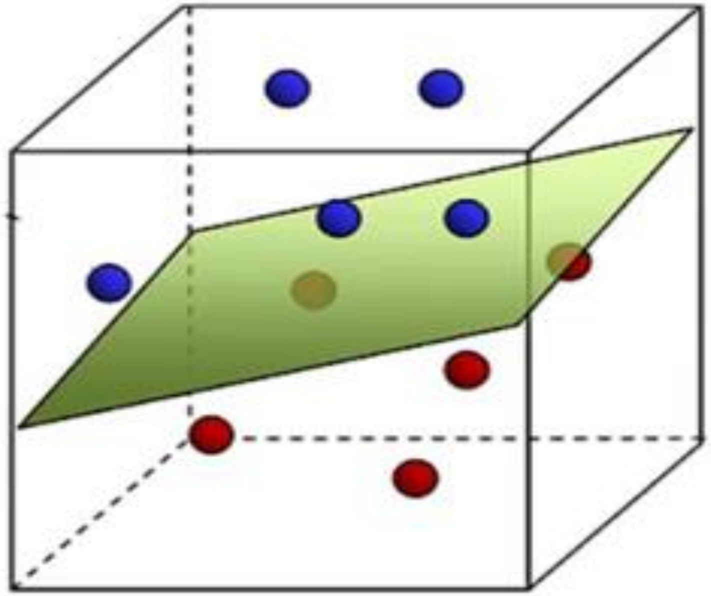

Not Linearly Separable
X_2
X_1

CONTD..

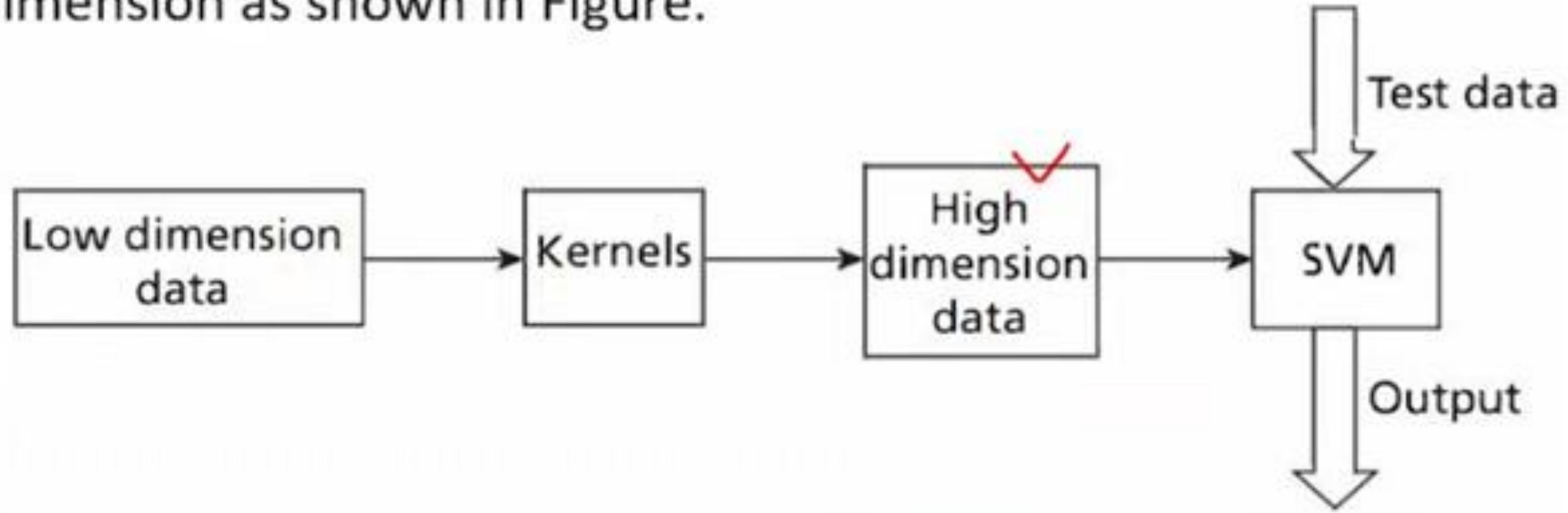
- So to **separate these data points**, we need **to add one more dimension.**
- For **linear data**, we have used **two dimensions x and y,** so for **non-linear data**, we will **add a third dimension z.**
- So ,here we use a **kernel function** to handle **non-linear separable data.**
- **Kernels** are used by **classification algorithms** to solve **non-linear classification problems.**
- The kernel function transforms the **data into a higher dimensional feature space** to make it possible to **perform the linear separation.**

- In machine learning applications, the data can be text, image, or video.
- So, there is a need to extract features from these data prior to classification.
- Hence, in the real world, many classification models are complex and mostly require non-linear hyperplanes.





- Kernels are a set of functions used to transform data from lower dimension to higher dimension and to manipulate data using dot product at higher dimensions.
- The use of kernels is to apply transformation to data and perform classification at the higher dimension as shown in Figure.



CONTD..

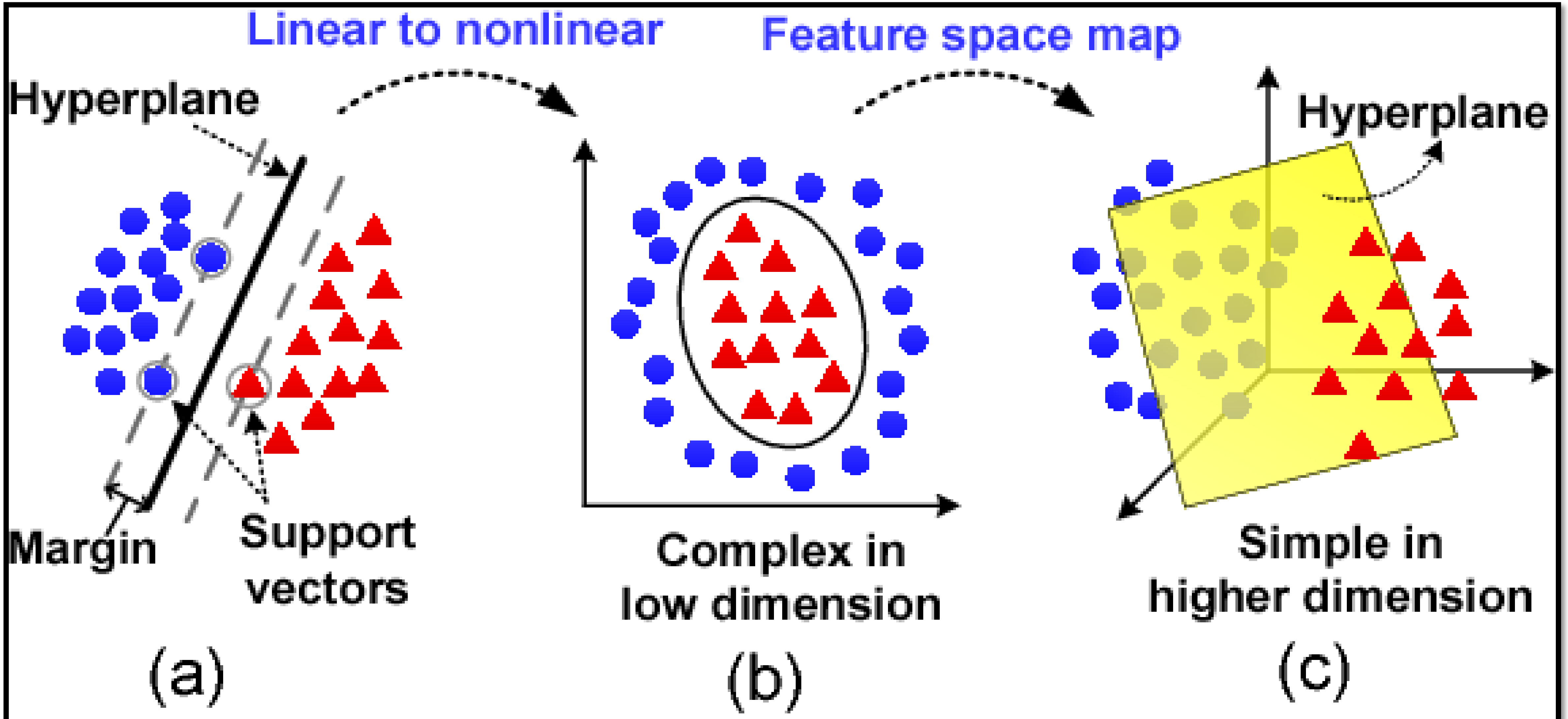
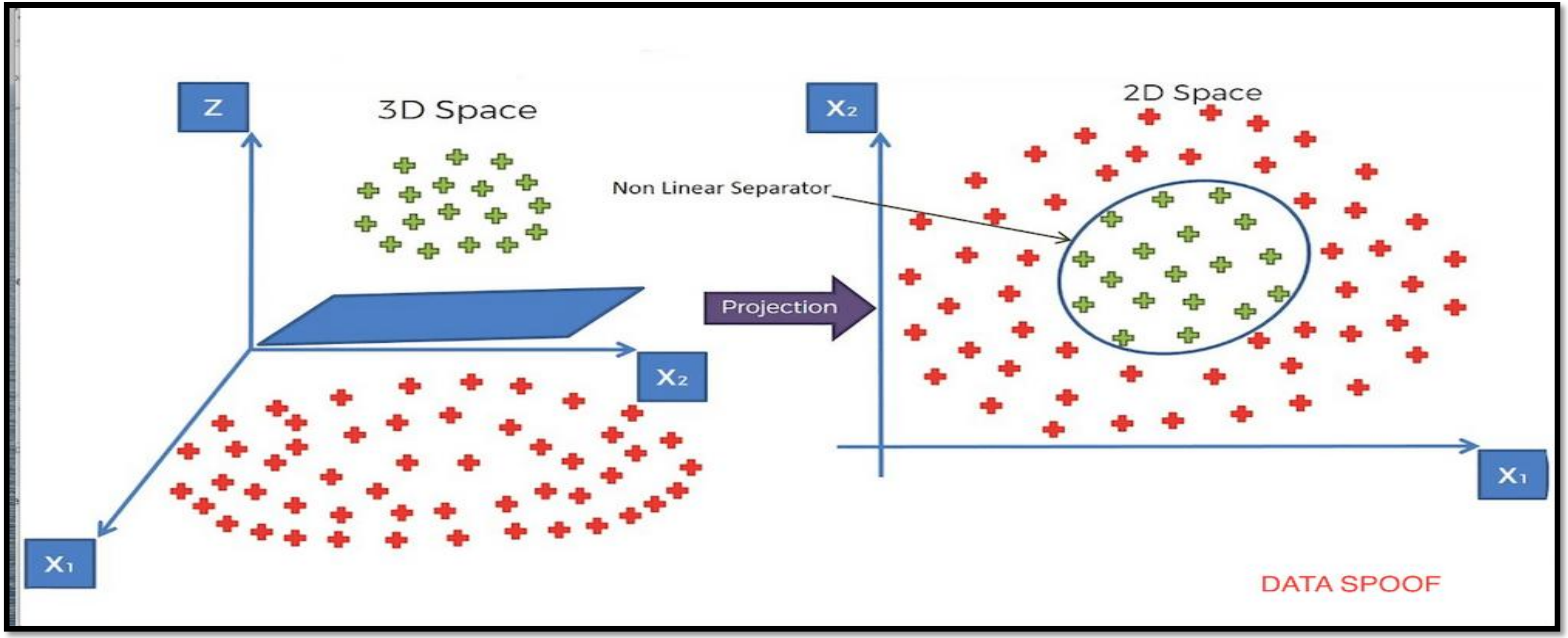


Fig. 1: SVM schematic diagram

EXAMPLE



Z
3D Space
X2
X1
Projection
X2
2D Space
Non Linear Separator
X1
DATA SPOOF

CONTD..

- he kernel function is a function that may be expressed as the **dot product of the mapping function** (kernel method) and looks like this, $\mathbf{K(x_i,x_j) = \text{Ø}(x_i) \cdot \text{Ø}(x_j)}$
- There are **different types of Kernels** are available to convert non linear to linear.

    1. **linear**: u'*v
    2. **polynomial**: (gamma*u'*v + coef0)^degree
    3. **radial basis** (RBF) : exp(-gamma*|u-v|^2)
    4. **sigmoid** : tanh(gamma*u'*v + coef0)

- **RBF is generally the most popular one.**

CONTD..

The mathematical formula behind RBF is:-

$$K(x, x') = e^{-\gamma ||x - x'|||^2}$$

Gamma is a scalar that defines how much influence a single training example (point) has.

EXAMPLE

In the **Given Dataset**, we **have 4 are positively labeled data sets and 4 are negatively labeled data sets.**

Suppose we are given the following positively labeled data points,

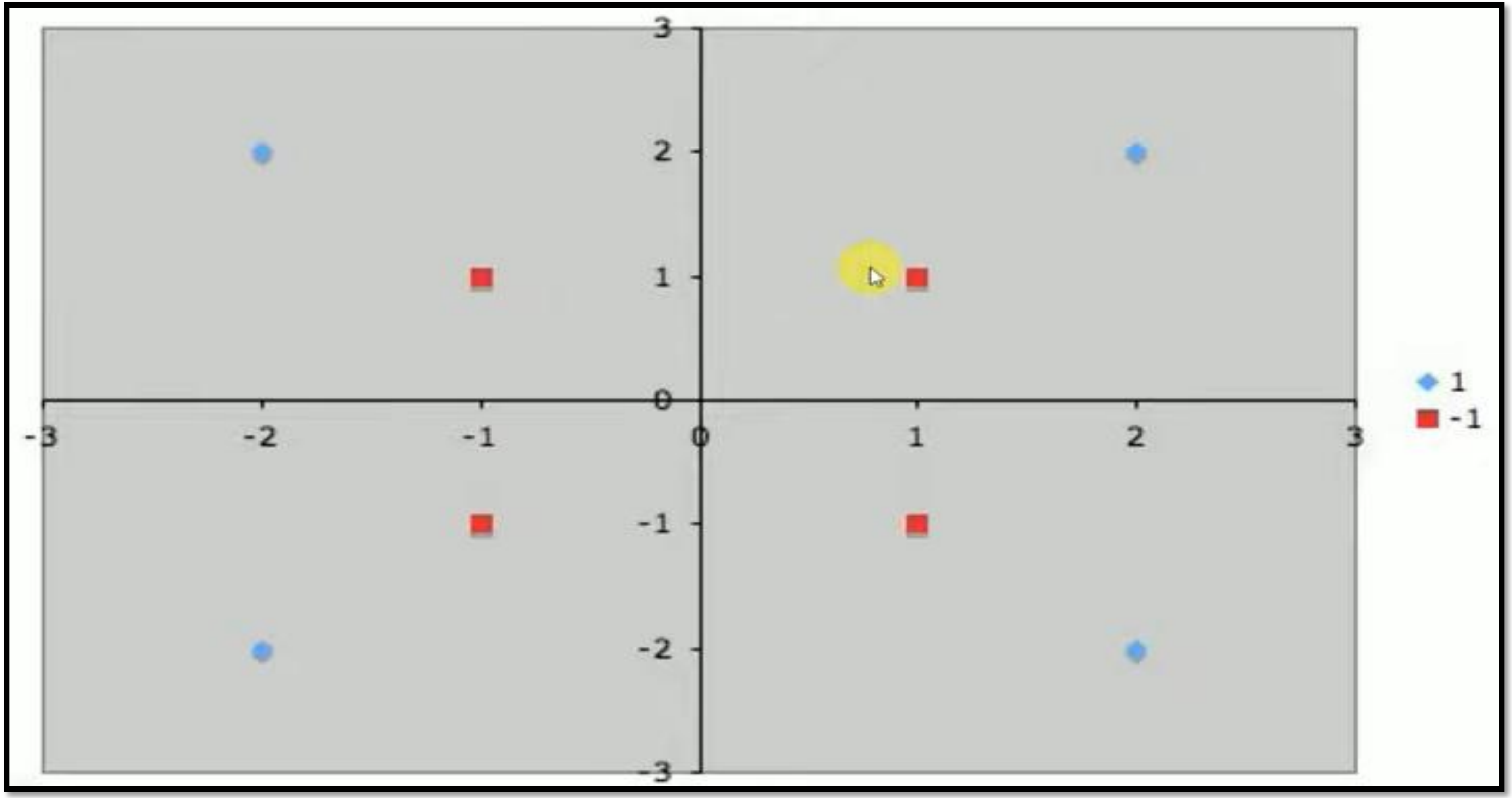
$$\left\{ \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ 2 \end{pmatrix} \right\}$$

and the following negatively labeled data points,

$$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$$

CONTD..

Next ,we need to plot the data points



Here our Goal is to separate Hyper plane that accurately separately two classes.

- Therefore, we must use a nonlinear SVM (that is, we need to convert data from one feature space to another.

$$\Phi_1\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{cases} \begin{pmatrix} 4 - x_2 + |x_1 - x_2| \\ 4 - x_1 + |x_1 - x_2| \end{pmatrix} & \text{if } \sqrt{x_1^2 + x_2^2} > 2 \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} & \text{otherwise} \end{cases}$$

Here our Goal is to separate Hyper plane that accurately separately two classes by using the Mapping Function RBF.

$$\Phi_1\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{cases} \begin{pmatrix} 4 - x_2 + |x_1 - x_2| \\ 4 - x_1 + |x_1 - x_2| \end{pmatrix} & \text{if } \sqrt{x_1^2 + x_2^2} > 2 \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} & \text{otherwise} \end{cases}$$

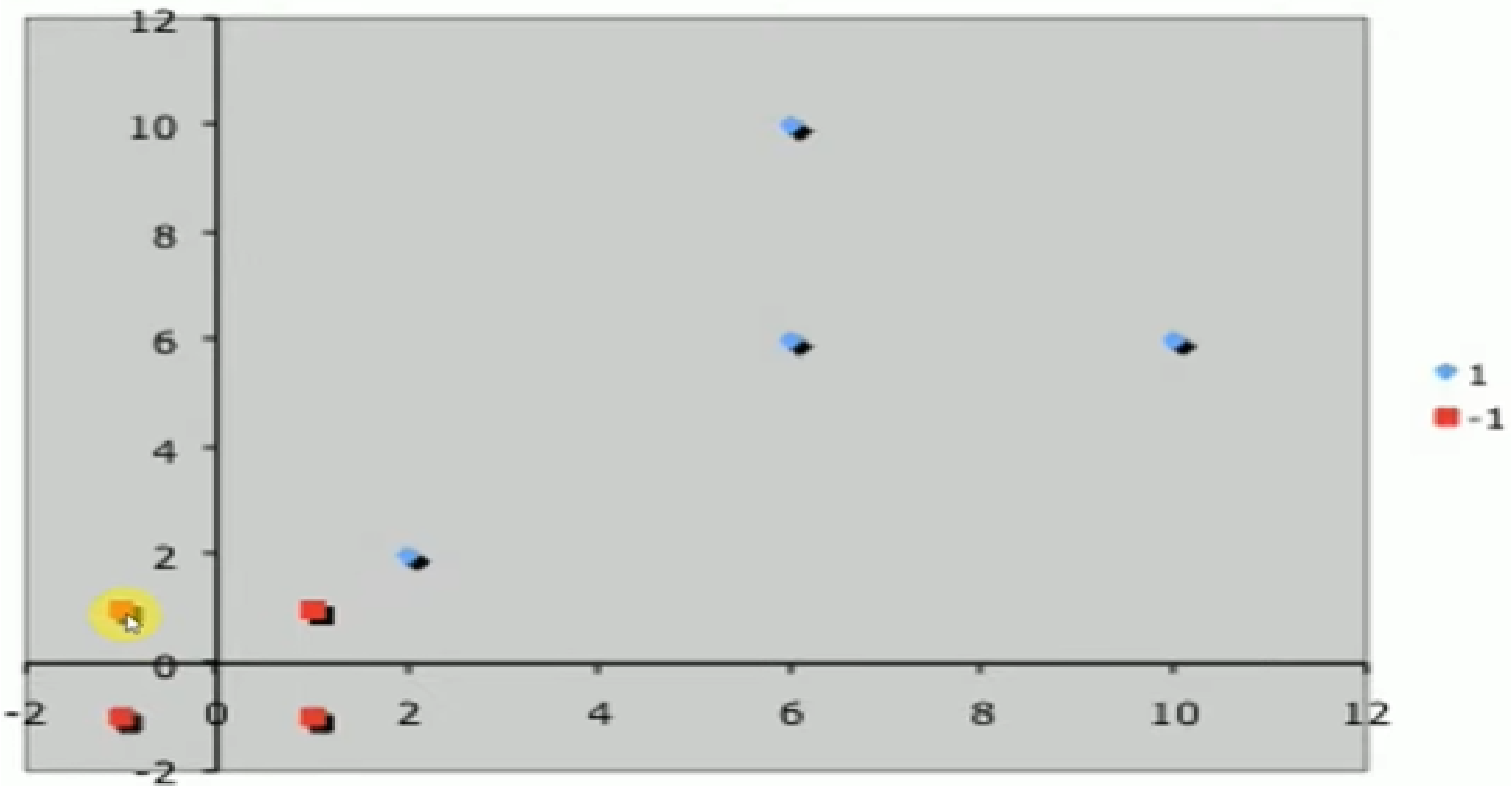
Positive Examples

$$\left\{\begin{pmatrix} 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ 2 \end{pmatrix}\right\} \rightarrow \left\{\begin{pmatrix} 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 10 \\ 6 \end{pmatrix}, \begin{pmatrix} 6 \\ 6 \end{pmatrix}, \begin{pmatrix} 6 \\ 10 \end{pmatrix}\right\}$$

Negative Examples

$$\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}\right\} \rightarrow \left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}\right\}$$

Next we can draw the Hyper plane by using updated Data Point



CONTD..

- Now we can easily identify the support vectors,

$$\left\{ s_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, s_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\}$$

- Each vector is augmented with a 1 as a bias input

$$\tilde{s_1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \qquad \tilde{s_2} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

We need to calculate the 3 Variables. ie, $\alpha 1$, $\alpha 2$.

Which will be used to calculate the Weight Vector.

$$\alpha_1 \tilde{s_1} \cdot \tilde{s_1} + \alpha_2 \tilde{s_2} \cdot \tilde{s_1} = -1$$

$$\alpha_1 \tilde{s_1} \cdot \tilde{s_2} + \alpha_2 \tilde{s_2} \cdot \tilde{s_2} = +1$$

$$\alpha_1 (1+1+1) + \alpha_2 (2+2+1) = -1$$

$$\alpha_1 (2+2+1) + \alpha_2 (4+4+1) = 1$$

$$\alpha_1 \begin{pmatrix}1\\1\\1\end{pmatrix}\begin{pmatrix}1\\1\\1\end{pmatrix} + \alpha_2 \begin{pmatrix}2\\2\\1\end{pmatrix}\begin{pmatrix}1\\1\\1\end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix}1\\1\\1\end{pmatrix}\begin{pmatrix}2\\2\\1\end{pmatrix} + \alpha_2 \begin{pmatrix}2\\2\\1\end{pmatrix}\begin{pmatrix}2\\2\\1\end{pmatrix} = 1$$

$$3\alpha_1 + 5\alpha_2 = -1$$

$$5\alpha_1 + 9\alpha_2 = 1$$

$$\alpha_1 = -7$$

$$\alpha_2 = 4$$

Once we get the 2 variables, after that we need to calculate the Weight Vector.

$$\tilde{w} = \sum_i \alpha_i \tilde{s}_i = -7\begin{pmatrix}1\\1\\1\end{pmatrix} + 4\begin{pmatrix}2\\2\\1\end{pmatrix}$$

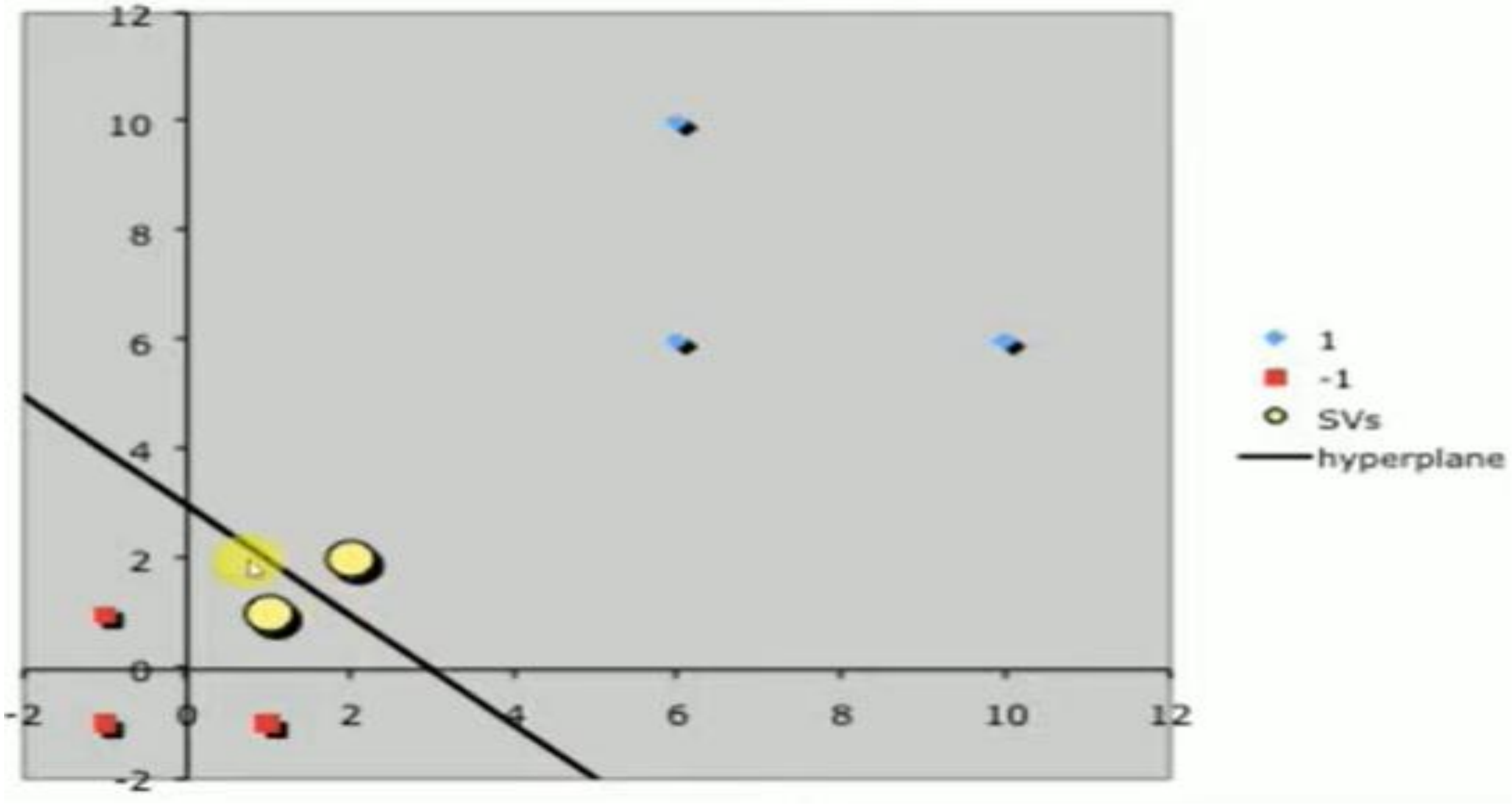
$$= \begin{pmatrix}1\\1\\-3\end{pmatrix}$$

- Finally, remembering that our vectors are augmented with a bias.
- We can equate the last entry in $\tilde{w}$ as the hyperplane offset b and write the separating
- Hyperplane equation $\boldsymbol{y} = \boldsymbol{wx} + \boldsymbol{b}$
- with $\boldsymbol{w} = \binom{1}{1}$ and $\boldsymbol{b} = -3$.

Here we get, the line is (1,0) ie, the line is parallel to Y- Axis.

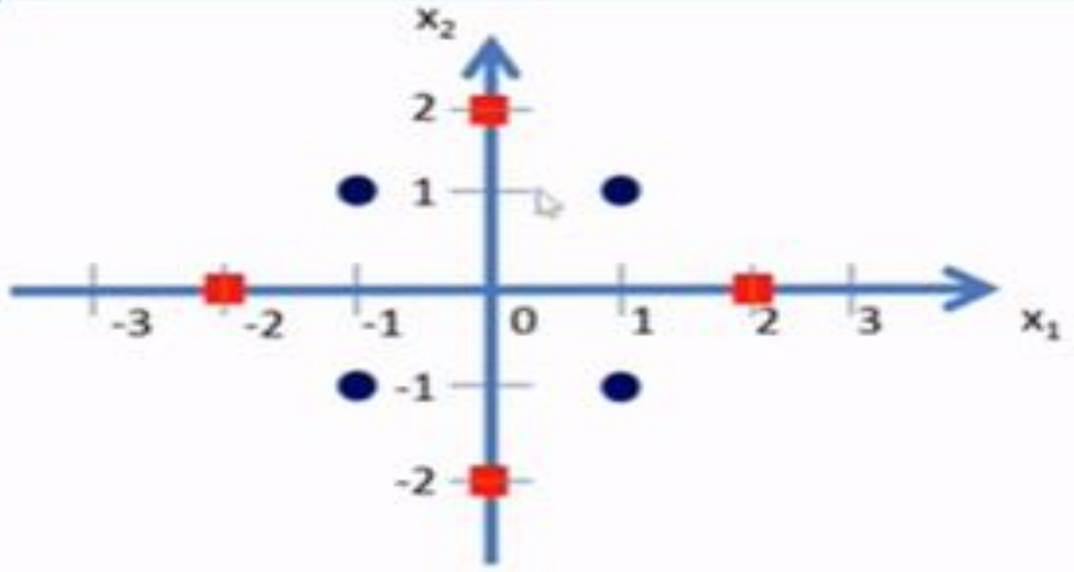
Suppose , If the Line is (0,1) ie, the line is parallel to X- Axis.

If the line is (1,1) , ie, the line is parallel to 45 Degrees.



EXAMPLE-2

In the **Given Dataset**, we **have 4 are positively labeled data sets and 4 are negatively labeled data sets.**



- Blue class vectors are: $\binom{1}{1}, \binom{-1}{1}, \binom{-1}{-1}, \binom{1}{-1}$
- Red class vectors are: $\binom{2}{0}, \binom{0}{2}, \binom{-2}{0}, \binom{0}{-2}$

Here our Goal is to separate Hyper plane that accurately separately two classes by using the Mapping Function RBF.

- $$\Phi\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{cases} \begin{pmatrix} 6 - x_1 + (x_1 - x_2)^2 \\ 6 - x_2 + (x_1 - x_2)^2 \end{pmatrix} & \text{if } \sqrt{x_1^2 + x_2^2} \geq 2 \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} & \text{otherwise} \end{cases}$$

- Now let us transform the blue and red calss vectors using the non-linear mapping function Φ .

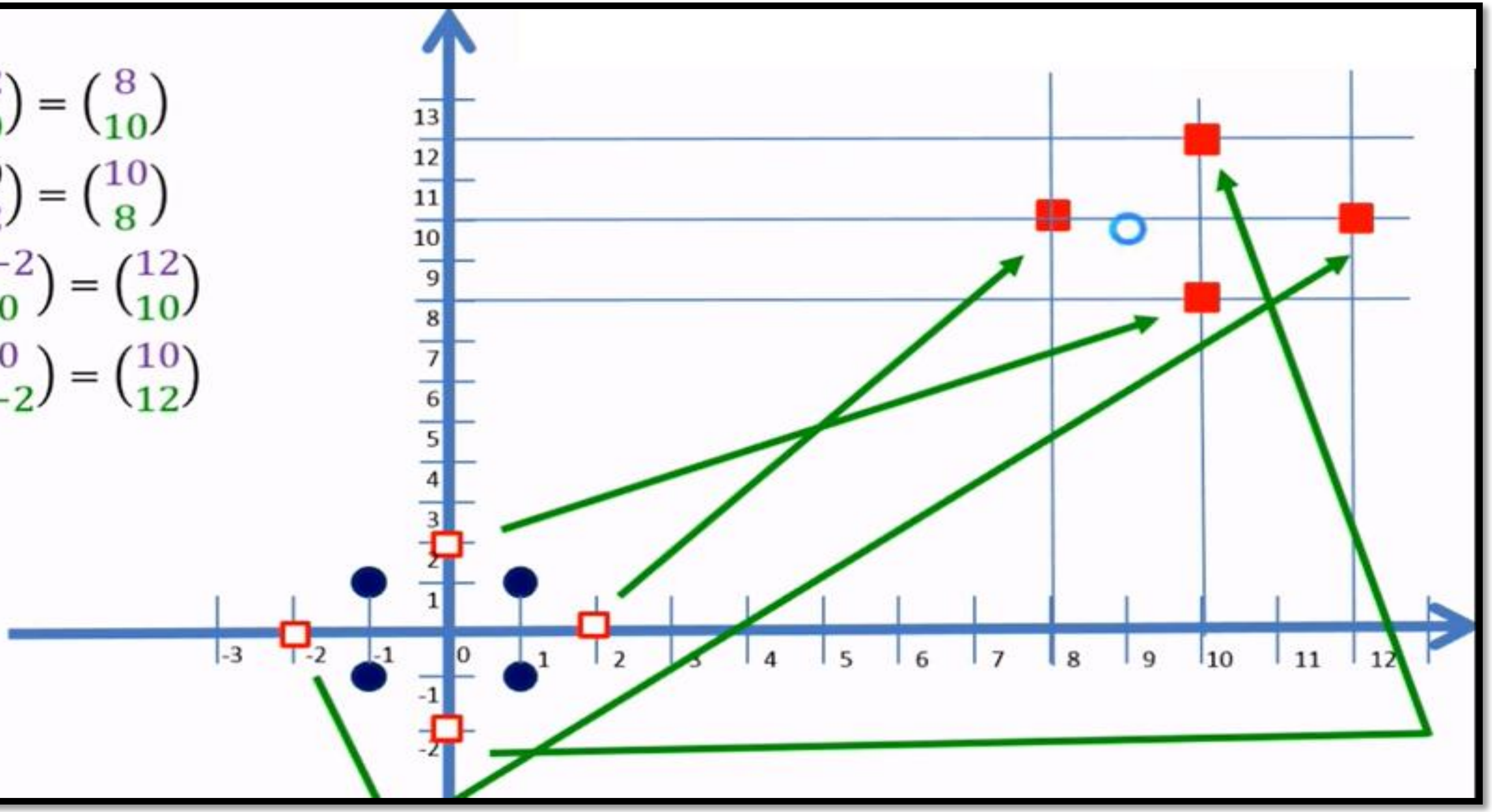
- $$\Phi\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{cases} \begin{pmatrix} 6 - x_1 + (x_1 - x_2)^2 \\ 6 - x_2 + (x_1 - x_2)^2 \end{pmatrix} & \text{if } \sqrt{x_1^2 + x_2^2} \geq 2 \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} & \text{otherwise} \end{cases}$$

- Blue class vectors are: $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ no change since $\sqrt{x_1^2 + x_2^2} < 2$ for all the vectors

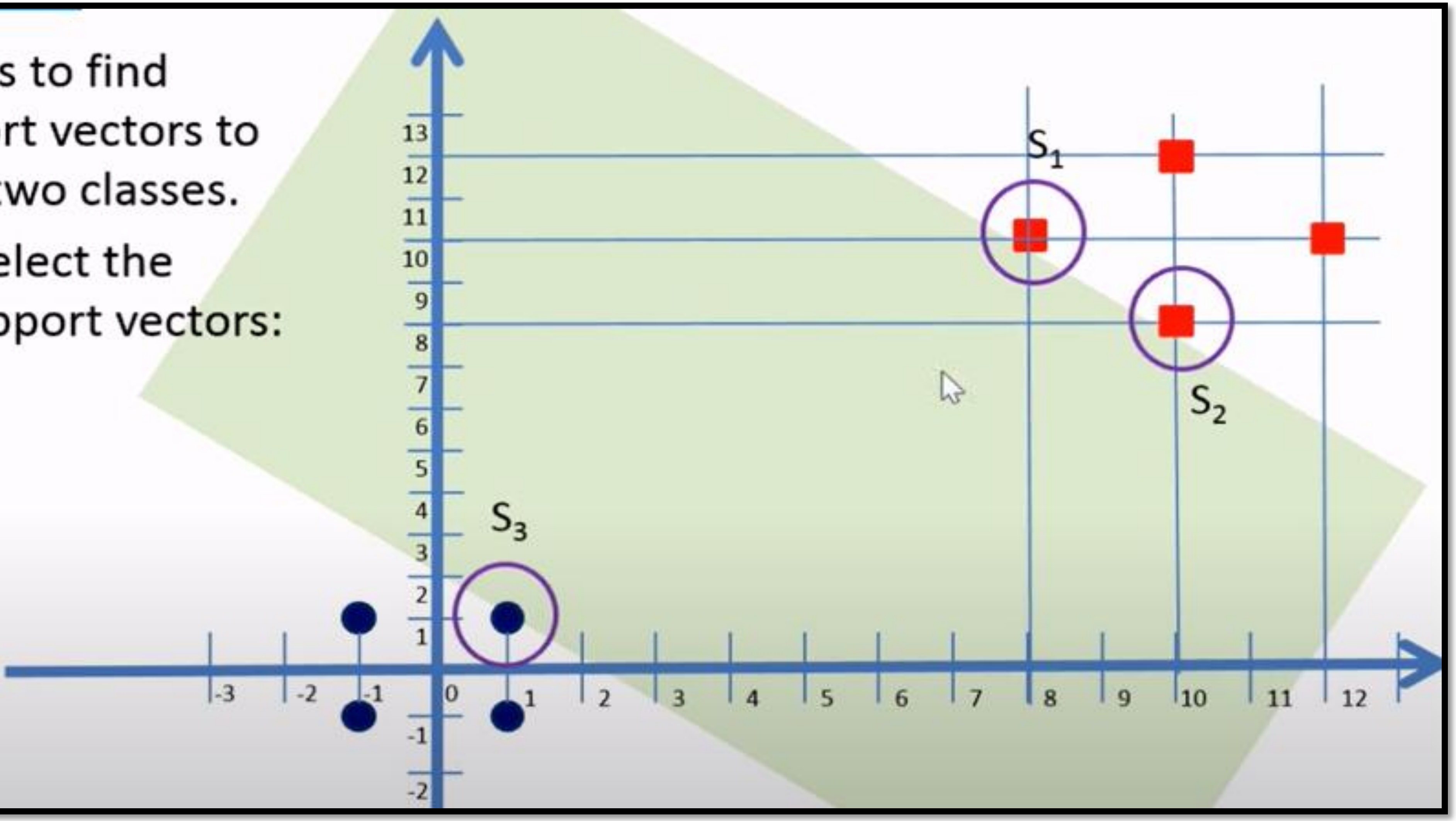
- $\Phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \begin{cases}\begin{pmatrix}6 - x_1 + (x_1 - x_2)^2\\ 6 - x_2 + (x_1 - x_2)^2\end{pmatrix} & \text{if } \sqrt{x_1^2 + x_2^2} \geq 2\\ \begin{pmatrix}x_1\\x_2\end{pmatrix} & \text{otherwise}\end{cases}$
- Let us take Red class vectors : $\begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}0\\2\end{pmatrix}, \begin{pmatrix}-2\\0\end{pmatrix}, \begin{pmatrix}0\\-2\end{pmatrix}$
- $\Phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \Phi\begin{pmatrix}2\\0\end{pmatrix} = \begin{pmatrix}6 - 2 + (2 - 0)^2\\ 6 - 0 + (2 - 0)^2\end{pmatrix} = \begin{pmatrix}8\\10\end{pmatrix}$
- $\Phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \Phi\begin{pmatrix}0\\2\end{pmatrix} = \begin{pmatrix}6 - 0 + (0 - 2)^2\\ 6 - 2 + (0 - 2)^2\end{pmatrix} = \begin{pmatrix}10\\8\end{pmatrix}$
- $\Phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \Phi\begin{pmatrix}-2\\0\end{pmatrix} = \begin{pmatrix}6 + 2 + (-2 - 0)^2\\ 6 - 0 + (-2 - 0)^2\end{pmatrix} = \begin{pmatrix}12\\10\end{pmatrix}$
- $\Phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \Phi\begin{pmatrix}0\\-2\end{pmatrix} = \begin{pmatrix}6 - 0 + (0 + 2)^2\\ 6 + 2 + (0 + 2)^2\end{pmatrix} = \begin{pmatrix}10\\12\end{pmatrix}$

Draw the Hyper Plane by using updated Data Points

- $\Phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \Phi\begin{pmatrix}2\\0\end{pmatrix} = \begin{pmatrix}8\\10\end{pmatrix}$
- $\Phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \Phi\begin{pmatrix}0\\2\end{pmatrix} = \begin{pmatrix}10\\8\end{pmatrix}$
- $\Phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \Phi\begin{pmatrix}-2\\0\end{pmatrix} = \begin{pmatrix}12\\10\end{pmatrix}$
- $\Phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \Phi\begin{pmatrix}0\\-2\end{pmatrix} = \begin{pmatrix}10\\12\end{pmatrix}$



- Now our task is to find suitable support vectors to classify these two classes.
- Here we will select the following 3 support vectors:
- $S_1 = \binom{8}{10}$,
- $S_2 = \binom{10}{8}$,
- and $S_3 = \binom{1}{1}$



Here we will add the Bias value

$$S_1 = \begin{pmatrix} 8 \\ 10 \end{pmatrix} \qquad \widetilde{S_1} = \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix}$$

$$S_2 = \begin{pmatrix} 10 \\ 8 \end{pmatrix} \qquad \widetilde{S_2} = \begin{pmatrix} 10 \\ 8 \\ 1 \end{pmatrix}$$

$$S_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \widetilde{S_3} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\alpha_1 \widetilde{S_1}.\widetilde{S_1} + \alpha_2 \widetilde{S_2}.\widetilde{S_1} + \alpha_3 \widetilde{S_3}.\widetilde{S_1} = +1 \ (+ve\ class)$$

$$\alpha_1 \widetilde{S_1}.\widetilde{S_2} + \alpha_2 \widetilde{S_2}.\widetilde{S_2} + \alpha_3 \widetilde{S_3}.\widetilde{S_2} = +1 \ (+ve\ class)$$

$$\alpha_1 \widetilde{S_1}.\widetilde{S_3} + \alpha_2 \widetilde{S_2}.\widetilde{S_3} + \alpha_3 \widetilde{S_3}.\widetilde{S_3} = -1 \ (-ve\ class)$$

- Let's substitute the values for $\widetilde{S_1}$, $\widetilde{S_2}$ and $\widetilde{S_3}$ in the above equations.

$$\widetilde{S_1} = \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix} \quad \widetilde{S_2} = \begin{pmatrix} 10 \\ 8 \\ 1 \end{pmatrix} \quad \widetilde{S_3} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\alpha_1 \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix} . \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 10 \\ 8 \\ 1 \end{pmatrix} . \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} . \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix} = +1$$

$$\alpha_1 \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix} . \begin{pmatrix} 10 \\ 8 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 10 \\ 8 \\ 1 \end{pmatrix} . \begin{pmatrix} 10 \\ 8 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} . \begin{pmatrix} 10 \\ 8 \\ 1 \end{pmatrix} = +1$$

$$\alpha_1 \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix} . \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 10 \\ 8 \\ 1 \end{pmatrix} . \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} . \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = -1$$

After simplification we get:

$$165\,\alpha_1 + 161\,\alpha_2 + 19\,\alpha_3 = -1$$

$$161\,\alpha_1 + 165\,\alpha_2 + 19\,\alpha_3 = -1$$

$$19\,\alpha_1 + 19\,\alpha_2 + 3\,\alpha_3 = +1$$

Simplifying the above 3 simultaneous equations we get: $\alpha_1 = \alpha_2 = 0.859$ and $\alpha_3 = -1.4219$.

- The hyper plane that discriminates the positive class from the negative class is given by:

$$\tilde{w} = \sum_i \alpha_i \tilde{S}_i$$

- Substituting the values we get:

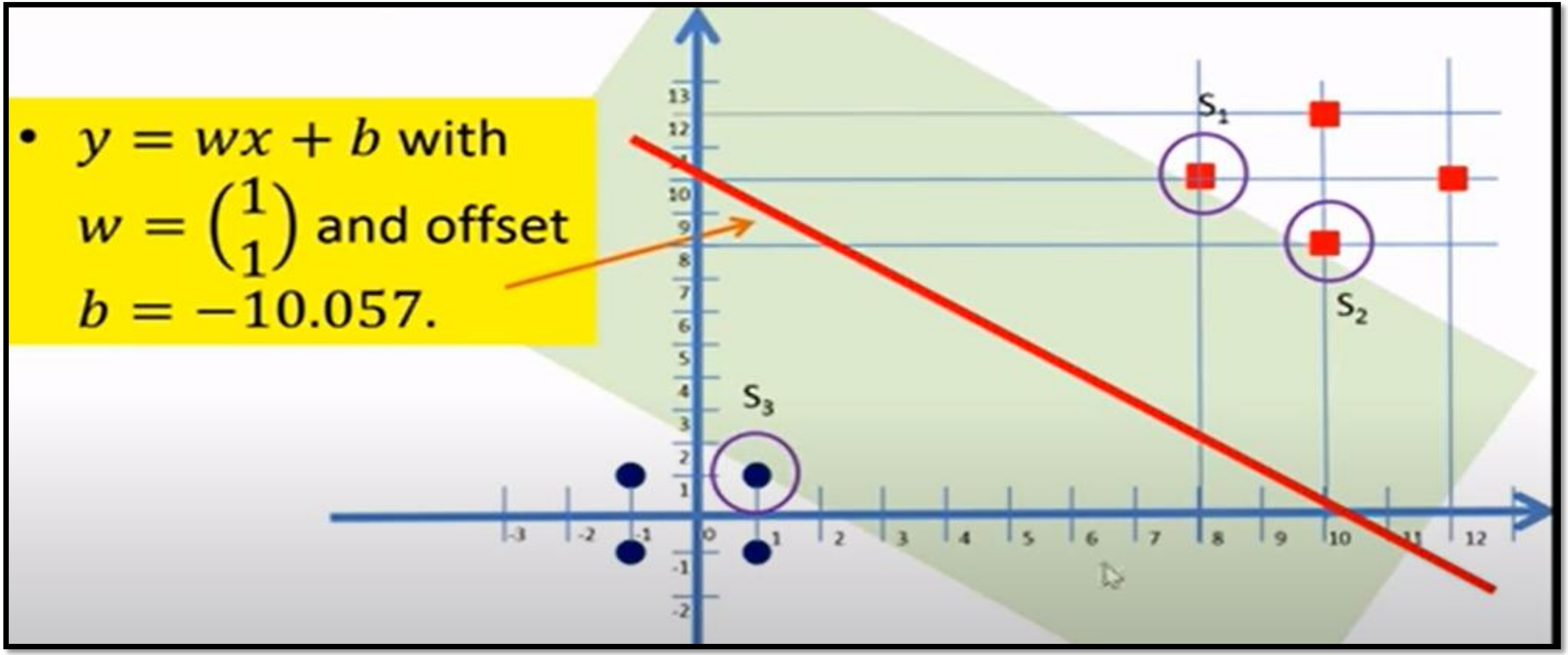
$$\tilde{w} = \alpha_1 \begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 10 \\ 8 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\tilde{w} = (0.0859).\begin{pmatrix} 8 \\ 10 \\ 1 \end{pmatrix} + (0.0859).\begin{pmatrix} 10 \\ 8 \\ 1 \end{pmatrix} + (-1.4219).\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.1243 \\ 0.1243 \\ -1.2501 \end{pmatrix}$$

- Our vectors are augmented with a bias.
- Hence we can equate the entry in $\widetilde{w}$ as the hyper plane with an offset b.
- Therefore the separating hyper plane equation

$$y = wx + b \text{ with } w = \begin{pmatrix} 0.125/0.125 \\ 0.125/0.125 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

and an offset $b = -\frac{1.250}{0.125} = -10.057.$


• $y = wx + b$ with
$w = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and offset
$b = -10.057.$
S_1
S_2
S_3